

47. Serial Sound Interface (SSIE)

47.1 Overview

This MCU has two serial sound interfaces (SSIE0 and SSIE1). Each SSIE supports monaural and time-division multiplexed (TDM) formats in addition to standard audio data formats such as I²S, left-justified, and right-justified.

Table 47.1 Features of SSIE

Item		Description
Number of channels		Two channels, SSIE0 and SSIE1
Transfer mode		• Master/slave • Transmission, reception, or transception (transception is only available on SSIE0)
Data format		• I²S format • Left-justified format • Right-justified format • Monaural format • TDM format
Serial data		• Fixed to MSB first • System word length: Selectable from among 8, 16, 24, 32, 48, 64, 128, or 256 bits • Data word length: Selectable from among 8, 16, 18, 20, 22, 24, or 32 bits • The level of padding bits is programmable. • Mute function
Bit clock (BCK)	In master mode	• Clock source: AUDIO_CLK • Frequency: Selectable from among 1/1, 1/2, 1/4, 1/6, 1/8, 1/12, 1/16, 1/24, 1/32, 1/48, 1/64, 1/96, or 1/128 of the AUDIO_CLK frequency • Continued supply or stopping while data transfer is halted is selectable.
	In master and slave modes	• Polarity is selectable as rising edges or falling edges.
LR clock (LRCK)		• Polarity is selectable as low or high. • Continued supply or stopping while data transfer is halted is selectable.
FIFO	Capacity	• Transmit FIFO: 4 bytes × 32 stages • Receive FIFO: 4 bytes × 32 stages
	Data alignment	• Alignment of data (left-justified or right-justified) in the FIFO is selectable.
Interrupts		• Data transfer error/idle state • Receive data full • Transmit data empty
Low power consumption function		• Module stop function • Master clock (MCK) supply stopping function

Table 47.2 lists and Figure 47.1 shows the terms and the definitions used in this chapter.

Table 47.2 Definitions of Terms

Term	Definition
Start trigger	First edge of the type specified by the SSICR.LRCKP bit of the signal on the SSILRCK pin after the start of data transfer
Frame boundary	Point where SSIE starts transferring the first data of a frame or the point where SSIE ends transferring the last data of the frame
Frame word length	Number of system words per frame (number of channels)
System word length	Number of transferred bits per channel
Data word length	Number of effective bits per channel
Master clock (MCK)	Clock for master mode input from the AUDIO_CLK pin

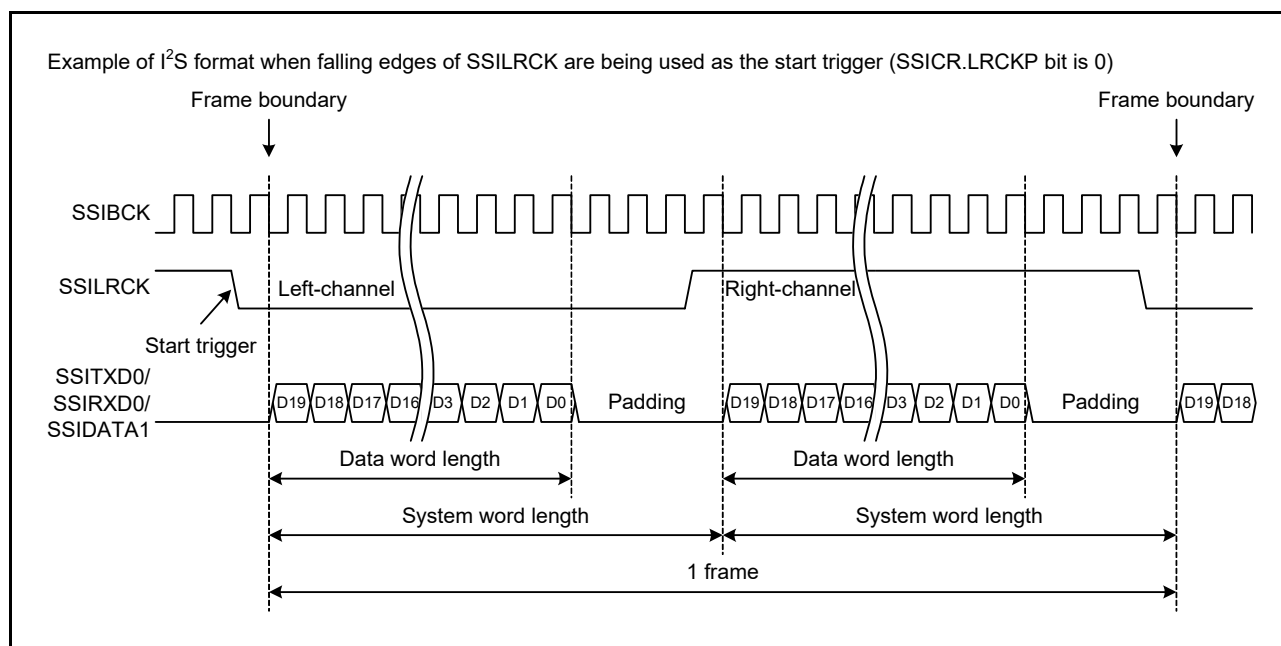


Figure 47.1 Definitions of Terms Related to Data Format

Figure 47.2 and Figure 47.3 show block diagrams of the SSIE0 and SSIE1, respectively.

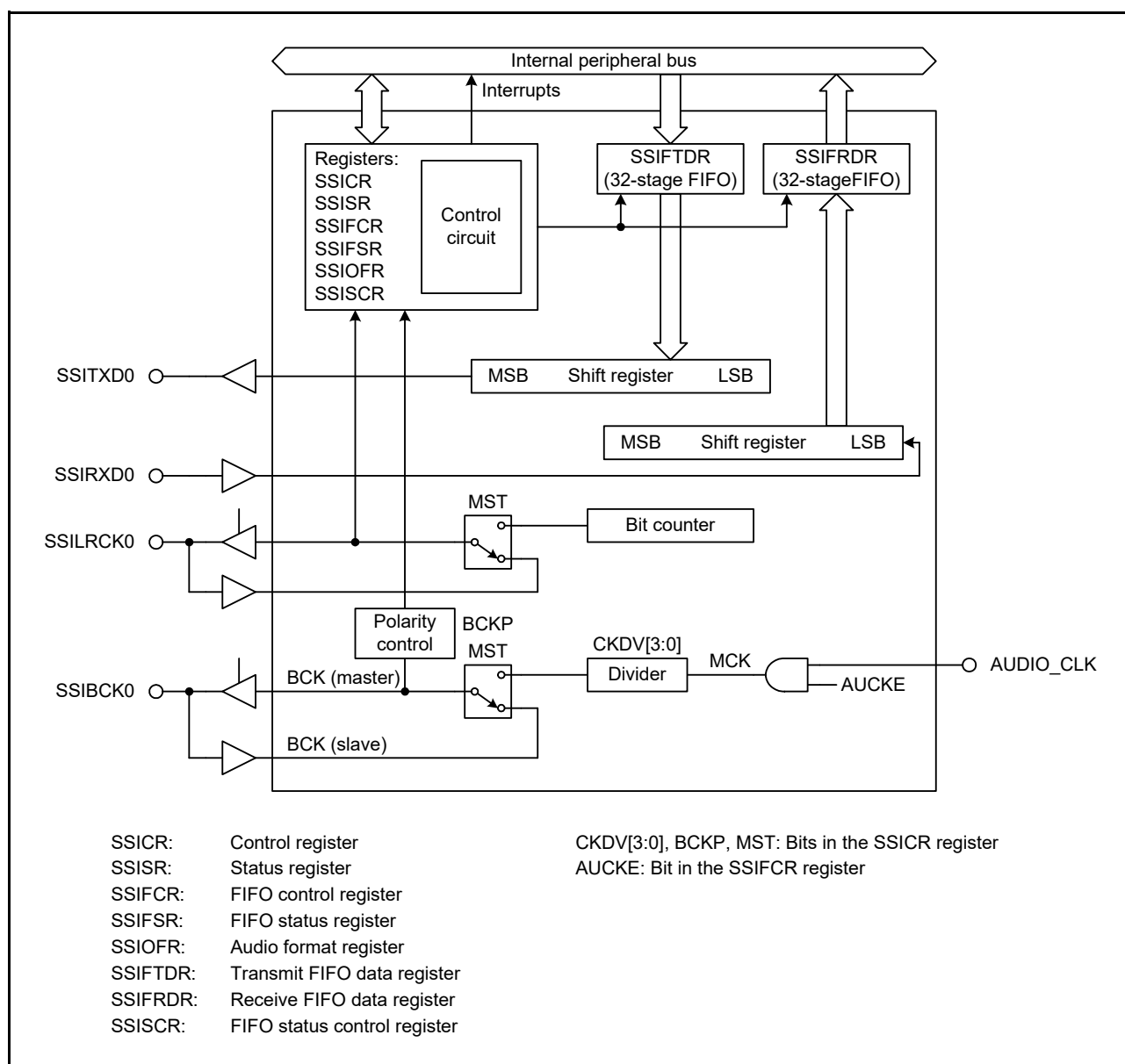


Figure 47.2 SSIE Block Diagram (SSIE0)

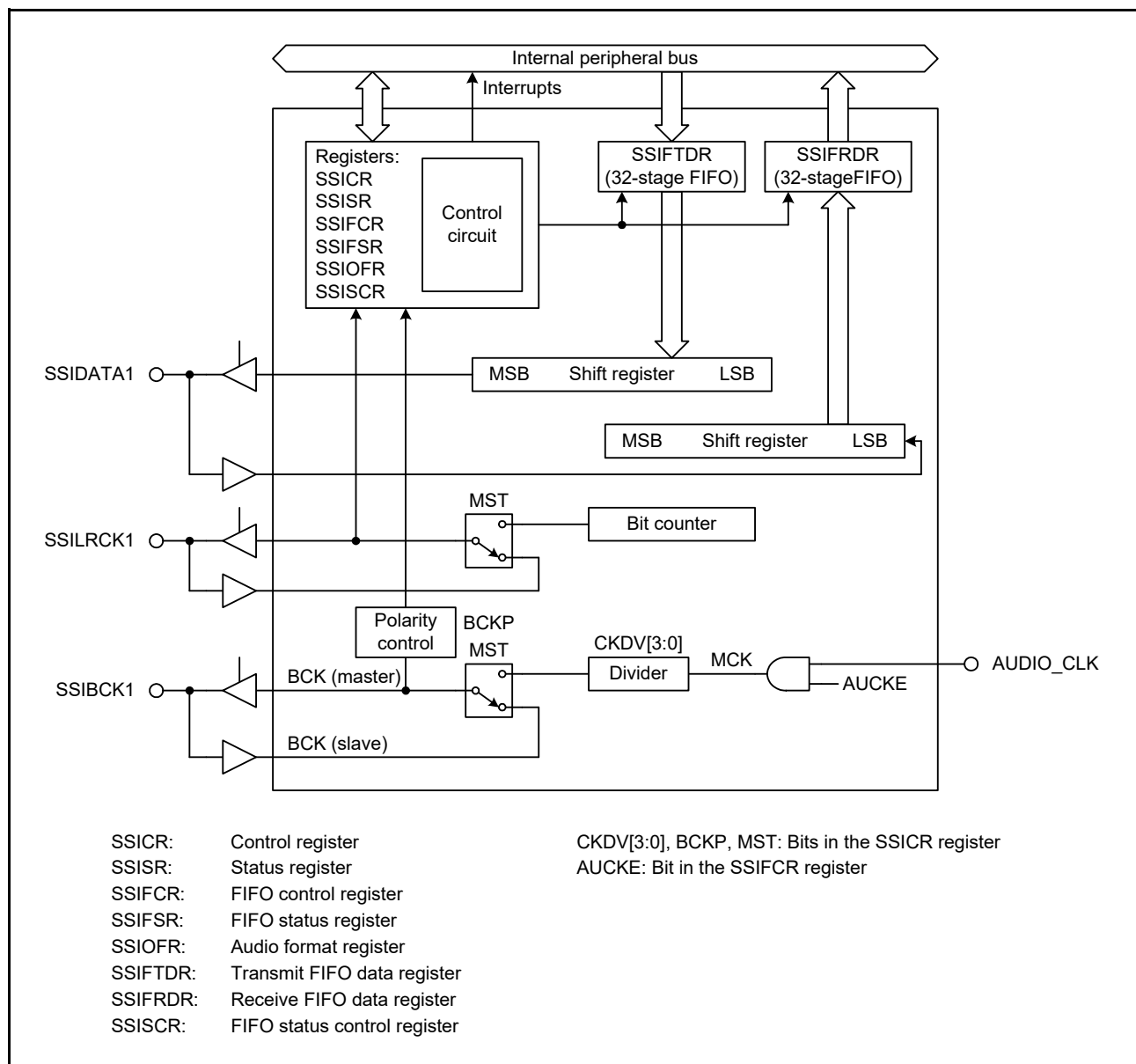


Figure 47.3 SSIE Block Diagram (SSIE1)

Table 47.3 lists the I/O pins of the SSIEs.

Table 47.3 I/O pins of SSIE

Pin Name	I/O	Function
AUDIO_CLK	Input	Audio clock input
SSIBCK0	I/O	Channel 0 bit-clock input/output
SSILRCK0	I/O	Channel 0 LR clock input/output
SSITXD0	Output	Channel 0 data output
SSIRXD0	Input	Channel 0 data input
SSIBCK1	I/O	Channel 1 bit-clock input/output
SSILRCK1	I/O	Channel 1 LR clock input/output
SSIDATA1	I/O	Channel 1 data input/output

47.2 Register Descriptions

47.2.1 Control Register (SSICR)

Address(es): SSIE0.SSICR 0008 A500h, SSIE1.SSICR 0008 A540h

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
—	—	TUIEN	TOIEN	RUIEN	ROIEN	IEN	—	FRM[1:0]		DWL[2:0]			SWL[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	MST	BCKP	LRCKP	SPDP	SDTA	PDTA	DEL		CKDV[3:0]			MUEN	—	TEN	REN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	REN	Reception Enable* ¹	0: Reception disabled 1: Reception enabled	R/W
b1	TEN	Transmission Enable* ¹	0: Transmission disabled 1: Transmission enabled	R/W
b2	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b3	MUEN	Mute	0: Unmute 1: Mute	R/W
b7 to b4	CKDV[3:0]	Bit Clock Division Ratio Select* ²	Set the division ratio to generate a bit clock from MCK. b7 b4 0 0 0 0: No division 0 0 0 1: Divided by 2 0 0 1 0: Divided by 4 0 0 1 1: Divided by 8 0 1 0 0: Divided by 16 0 1 0 1: Divided by 32 0 1 1 0: Divided by 64 0 1 1 1: Divided by 128 1 0 0 0: Divided by 6 1 0 0 1: Divided by 12 1 0 1 0: Divided by 24 1 0 1 1: Divided by 48 1 1 0 0: Divided by 96 Settings other than above are prohibited.	R/W
b8	DEL	Serial Data Delay Select* ²	Set the delay between the edges of SSILRCK and the input and output of data. 0: Delay of 1 cycle of SSIBCK 1: No delay	R/W
b9	PDTA	Data Alignment Select* ²	Set the data alignment of the SSIFTDR and SSIFRDR registers. 0: Left-justified 1: Right-justified	R/W
b10	SDTA	Serial Data Alignment Select* ²	0: I ² S and left-justified format (data bits are transmitted and received before the padding bits) 1: Right-justified format (data bits are transmitted and received after the padding bits)	R/W
b11	SPDP	Padding Data Select* ²	0: Padding bits are at the low level. 1: Padding bits are at the high level.	R/W
b12	LRCKP	LR Clock Polarity Select* ²	0: SSILRCK is low for left channel. The frame boundary is a falling edge of SSILRCK. 1: SSILRCK is high for left channel. The frame boundary is a rising edge of SSILRCK.	R/W

Bit	Symbol	Bit Name	Description	R/W
b13	BCKP	Bit Clock Polarity Select*2	0: Output signals change at a falling edge of bit clock and input signals are sampled at a rising edge of bit clock. 1: Output signals change at a rising edge of bit clock and input signals are sampled at a falling edge of bit clock.	R/W
b14	MST	Master Mode*2	0: Slave mode 1: Master mode	R/W
b15	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b18 to b16	SWL[2:0]	System Word Length Select*2	b18 b16 0 0 0: 8 bits 0 0 1: 16 bits 0 1 0: 24 bits 0 1 1: 32 bits 1 0 0: 48 bits 1 0 1: 64 bits 1 1 0: 128 bits 1 1 1: 256 bits	R/W
b21 to b19	DWL[2:0]	Data Word Length Select*2	b21 b19 0 0 0: 8 bits 0 0 1: 16 bits 0 1 0: 18 bits 0 1 1: 20 bits 1 0 0: 22 bits 1 0 1: 24 bits 1 1 0: 32 bits 1 1 1: Setting prohibited.	R/W
b23, b22	FRM[1:0]	Frame Word Length Select*2	Set a number of system words per a frame. b23 b22 0 0: 2 system words (I ² S, left-justified, or right-justified) or 1 system word (monaural) 0 1: 4 system words (TDM format only) 1 0: 6 system words (TDM format only) 1 1: 8 system words (TDM format only)	R/W
b24	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b25	I IEN	Idle State Interrupt Enable	0: Idle state interrupt disabled 1: Idle state interrupt enabled	R/W
b26	ROIEN	Receive Overflow Interrupt Enable	0: Receive overflow interrupt disabled 1: Receive overflow interrupt enabled	R/W
b27	RUIEN	Receive Underflow Interrupt Enable	0: Receive underflow interrupt disabled 1: Receive underflow interrupt enabled	R/W
b28	TOIEN	Transmit Overflow Interrupt Enable	0: Transmit overflow interrupt disabled 1: Transmit overflow interrupt enabled	R/W
b29	TUIEN	Transmit Underflow Interrupt Enable	0: Transmit underflow interrupt disabled 1: Transmit underflow interrupt enabled	R/W
b31, b30	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. After the TEN or REN bit is rewritten, confirm that the SSISR.IIRQ flag is in the desired status (when transmission or reception is enabled, confirm that the IIRQ flag is 0; when transmission and reception are disabled, confirm that the IIRQ flag is 1).

Note 2. Rewrite these bits while SSIE is in idle state (SSISR.IIRQ flag = 1). If the value of these bits is changed while data transfer is in progress, the operation after the change is not guaranteed.

TEN Bit (Transmission Enable) and REN Bit (Reception Enable)

These bits enable/disable transmission and reception. When one of these bits is set to 1, the given type of data transfer starts in synchronization with a start trigger of the SSILRCK signal. When the bit is set to 0, the data transfer stops at the next frame boundary. To use an SSIE for both transmission and reception, set both the TEN and REN bits to 1 simultaneously. To stop the data transfer, set both the TEN and REN bits to 0 simultaneously.

To switch among transmission, reception, and transection, do so in the idle state.

MUEN Bit (Mute)

This bit is used to mute audio output by driving the SSITXD0/SSIDATA1 pin low. When this bit is changed during a frame, the setting takes effect at the next frame boundary.

The MUEN bit only controls fixing of the SSITXD0/SSIDATA1 pin. Status flags are set and interrupt signals are generated regardless of the setting of the MUEN bit.

Changing the value of this bit must be performed after setting the data format to be used.

CKDV[3:0] Bits (Bit Clock Division Ratio Select)

These bits specify the divisor for the generation of a bit clock (BCK) from the master clock (MCK). This bit is disabled in slave mode.

Set these bits while the SSIFCR.AUCKE bit is 0.

DEL Bit (Serial Data Delay Select)

When this bit is 0, a delay of one cycle of the bit clock is inserted between edges of SSILRCK0 and SSITXD0 and SSIRXD0 or edges of SSILRCK1 and SSIDATA1. When 1, no delay is inserted.

Set this bit to 0 for the I²S format and to 1 for the left-justified or right-justified format. Set this bit in conformance with the other party device for the monaural and TDM formats. When the monaural format is used, setting of this bit changes the high width of SSILRCK.

PDTA Bit (Data Alignment Select)

This bit specifies the alignment of the effective bits in the SSIFTDR and SSIFRDR registers. This bit is valid when the data word length 18 to 24 bits (the DWL[2:0] bits = 010b to 101b). Set this bit to 0, when the other data word length is selected.

Figure 47.4 shows the data alignment of the SSIFTDR and SSIFRDR registers.

DWL[2:0] bits	PDTA bit = 0 (left-justified)	PDTA bit = 1 (right-justified)
000b (8 bits)		Setting prohibited
001b (16 bits)		Setting prohibited
010b (18 bits)		
011b (20 bits)		
100b (22 bits)		
101b (24 bits)		
110b (32 bits)		Setting prohibited

Figure 47.4 Data Alignment in the SSIFTDR and SSIFRDR Registers

SDTA Bit (Serial Data Alignment Select)

This bit specifies the order of data bits and padding bits for transection in system words. SSIE transmits and receives data bits first and then padding bits when this bit is 0 and padding bits first and then data bits when this bit is 1. This bit is invalid for data format without padding bits.

LRCKP Bit (LR Clock Polarity Select)

This bit specifies the initial status and polarity of the SSILRCK pin. Table 47.4 lists the settings of this bit and the corresponding states of the LR clock. SSIE uses the SSILRCK signal as a start trigger in slave mode. Set this bit to 0 when the I²S format is selected and 1 when the other data formats are selected.

Table 47.4 The Value of the LRCKP Bit and Responses to the States of the LR clock

LRCKP bit	Initial State	Left Channel	Right Channel	Frame Boundary
0	High	Low	High	Falling edge
1	Low	High	Low	Rising edge

BCKP Bit (Bit Clock Polarity Select)

This bit specifies the polarity of the bit clock. Table 47.5 lists the settings of this bit and the corresponding timings of the respective signals.

Set this bit while the SSIFCR.AUCKE bit is 0.

Table 47.5 The Value of the BCKP bit and the Bit Clock Polarity

BCKP bit	Output of SSILRCK, SSITXD0, and SSIDATA1	Input of SSILRCK, SSIRXD0, and SSIDATA1
0	The signals change on falling edges of SSIBCK.	The signals are sampled on rising edges of SSIBCK.
1	The signals change on rising edges of SSIBCK.	The signals are sampled on falling edges of SSIBCK.

SWL[2:0] Bits (System Word Length Select)

These bits specify the number of bits in a system word. The setting of these bits determines the frequency ratio of the bit clock to the LR clock. Then number of the padding bits is obtained from the difference in the SWL[2:0] and DWL[2:0] bits. Refer to Table 47.8 for details.

DWL[2:0] Bits (Data Word Length Select)

These bits specify the number of bits of valid data in a system word. The data word length must not exceed the system word length. For details, refer to Table 47.8.

FRM[1:0] Bits (Frame Word Length Select)

These bits specify the number of system words in a frame.

Set these bits to 00b when the I²S, left-justified, right-justified, or monaural format is selected. Do not set these bits to 00b when the TDM format is selected.

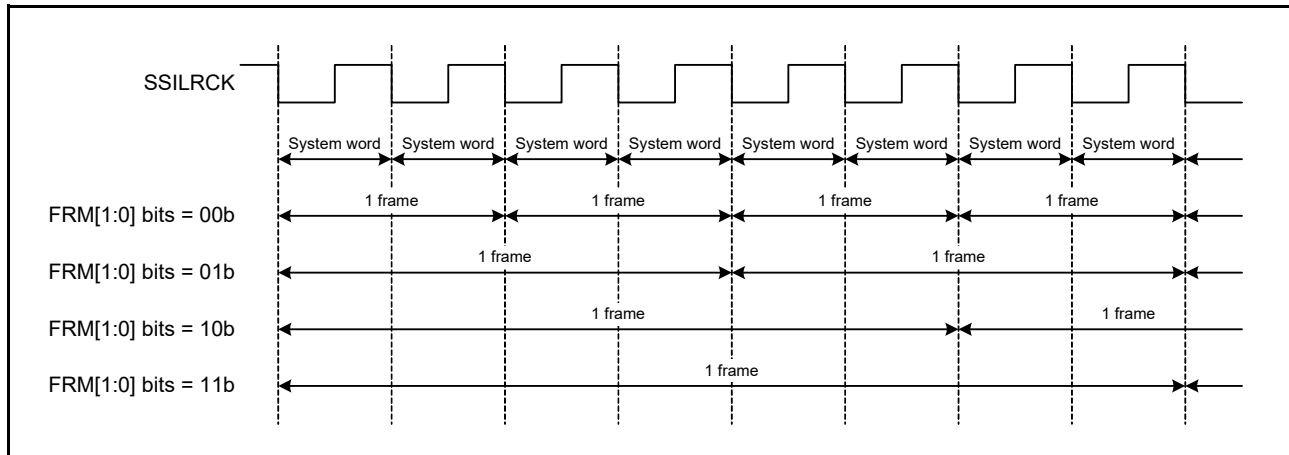


Figure 47.5 Frame Word Length

IEN Bit (Idle State Interrupt Enable)

This bit enables or disables the idle state interrupts.

While this bit is 1, an interrupt request is output when the SSISR.IIRQ flag becomes 1. An interrupt request is also output when this bit is changed from 0 to 1 while the IIRQ flag is 1.

ROEN Bit (Receive Overflow Interrupt Enable)

This bit enables or disables interrupts receive overflow interrupts.

While this bit is 1, an interrupt request is output when the SSISR.ROIRQ flag becomes 1. An interrupt request is also output when this bit is changed from 0 to 1 while the ROIRQ flag is 1.

RUIEN Bit (Receive Underflow Interrupt Enable)

This bit enables/disables the receive underflow interrupts.

While this bit is 1, an interrupt request is output when the SSISR.RUIRQ flag becomes 1. An interrupt request is also output when this bit is changed from 0 to 1 while the RUIRQ flag is 1.

TOEN Bit (Transmit Overflow Interrupt Enable)

This bit enables/disables the transmit overflow interrupts.

While this bit is 1, an interrupt request is output when the SSISR.TOIRQ flag becomes 1. An interrupt request is also output when this bit is changed from 0 to 1 while the TOIRQ flag is 1.

TUIEN Bit (Transmit Underflow Interrupt Enable)

This bit enables/disables the transmit underflow interrupts.

While this bit is 1, an interrupt request is output when the SSISR.TUIRQ flag becomes 1. An interrupt request is also output when this bit is changed from 0 to 1 while the TUIRQ flag is 1.

47.2.2 Status Register (SSISR)

Address(es): SSIE0.SSISR 0008 A504h, SSIE1.SSISR 0008 A544h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	TUIRQ	TOIRQ	RUIRQ	ROIRQ	IIRQ	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b24 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b25	IIRQ	Idle State Flag	0: Transfer state 1: Idle state	R
b26	ROIRQ	Receive Overflow Flag	0: No receive overflows occurred. 1: A receive overflow has occurred.	R/W
b27	RUIRQ	Receive Underflow Flag	0: No receive underflows occurred. 1: A receive underflow has occurred.	R/W
b28	TOIRQ	Transmit Overflow Flag	0: No transmit overflows occurred. 1: A transmit overflow has occurred.	R/W
b29	TUIRQ	Transmit Underflow Flag	0: No transmit underflows occurred. 1: A transmit underflow has occurred.	R/W
b31, b30	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

This register indicates the SSIE status.

IIRQ Flag (Idle State Flag)

This flag indicates whether SSIE is in the idle state or transfer state.

[Setting condition]

- When a transmission of a frame is completed after changing the SSICR.TEN bit from 1 to 0.
- When a reception of a frame is completed after changing the SSICR.REN bit from 1 to 0.
- When a transmission of a frame is completed after changing the SSICR.TEN and SSICR.REN bits from 1 to 0.

[Clearing condition]

- When a start trigger is generated by writing a transmit data in the SSIFTDR register, while the SSICR.TEN bit is 1.
- When a start trigger is detected while the SSICR.REN bit is 1.
- When a start trigger is generated by writing a transmit data to the SSIFTDR register, while the SSICR.TEN and SSICR.REN bits are 1.

ROIRQ Flag (Receive Overflow Flag)

This flag indicates a receive overflow. This flag indicates that the data processing of the received data is delayed. The received data is not transferred to the receive FIFO if a receive overflow occurs. This flag is not cleared by resetting a receive FIFO.

[Priority order for setting and clearing]

Setting is prioritized.

[Setting condition]

- When a reception of the next data is completed while the receive FIFO is full.

[Clearing condition]

- When 0 is written to this flag after confirming this flag is 1.
- When the SSICR.REN bit is changed from 0 to 1.

RUIRQ Flag (Receive Underflow Flag)

This flag indicates a receive underflow. This flag indicates that the SSIFRDR register is read while the receive FIFO is empty. Data read from the SSIFRDR register is invalid if a receive underflow occurs. This flag is not cleared by resetting a receive FIFO. Reading the SSIFRDR register while resetting the receive FIFO does not lead to setting of this flag.

[Priority order for setting and clearing]

Setting is prioritized.

[Setting condition]

- When the SSIFRDR register is read while the receive FIFO is empty.

[Clearing condition]

- When 0 is written to this flag after confirming this flag is 1.
- When the SSICR.REN bit is changed from 0 to 1.

TOIRQ Flag (Transmit Overflow Flag)

This flag indicates a transmit overflow. This flag indicates that a data is written to the SSIFTDR register while the transmit FIFO is full. The written data is discarded. This flag is not cleared by resetting a transmit FIFO.

[Priority order for setting and clearing]

Setting is prioritized.

[Setting condition]

- When a data is written to the SSIFTDR register while the transmit FIFO is full.

[Clearing condition]

- When 0 is written to this flag after confirming this flag is 1.
- When the SSICR.TEN bit is changed from 0 to 1.

TUIRQ Flag (Transmit Underflow Flag)

This flag indicates a transmit underflow. This flag indicates that supply of the transmit data was not enough to the transmission of the frame. The SSITXD0 and SSIDATA1 pins become low if a transmit underflow occurs, and remain low even if the flag is cleared. This flag is not cleared by resetting a transmit FIFO.

[Priority order for setting and clearing]

Setting is prioritized.

[Setting condition]

- When a data transfer to the transmit shift register occurs while the transmit FIFO is empty.

[Clearing condition]

- When 0 is written to this flag after confirming this flag is 1.
- When the SSICR.TEN bit is changed from 0 to 1.

47.2.3 FIFO Control Register (SSIFCR)

Address(es): SSIE0.SSIFCR 0008 A510h, SSIE1.SSIFCR 0008 A550h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	AUCKE	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SSIRST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	BSW	—	—	—	—	—	—	—	TIE	RIE	TFRST	RFRST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	RFRST	Receive FIFO Reset* ¹	0: The receive FIFO is released from the reset state. 1: The receive FIFO is reset.	R/W
b1	TFRST	Transmit FIFO Reset* ¹	0: The transmit FIFO is released from the reset state. 1: The transmit FIFO is reset.	R/W
b2	RIE	Receive Data Full Interrupt Enable	0: Receive data full interrupt is disabled. 1: Receive data full interrupt is enabled.	R/W
b3	TIE	Transmit Data Empty Interrupt Enable	0: Transmit data empty interrupt is disabled. 1: Transmit data empty interrupt is enabled.	R/W
b10 to b4	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b11	BSW	Byte Swap* ¹	0: Byte swapping is disabled 1: Byte swapping is enabled	R/W
b15 to b12	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b16	SSIRST	Software Reset	0: The software reset is released. 1: A software reset is applied.	R/W
b30 to b17	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b31	AUCKE	MCK Supply Enable* ¹	0: MCK is stopped. 1: MCK is supplied.	R/W

Note 1. Rewrite these bits while SSIE is in idle state (SSISR.IIRQ flag = 1). If the value of these bits is changed while data transfer is in progress, the operation after the change is not guaranteed.

RFRST Bit (Receive FIFO Reset)

This bit is used to reset the receive FIFO. Writing 1 to this bit resets the receive FIFO and initializes the RDC[5:0] bits and RDF flag in the SSIFSR register. Since this bit is not automatically cleared, write 0 to this bit to release from the reset state. After writing 0 to this bit, be sure to check that this bit is 0 before starting the next procedure.

This bit is subject to the software reset by the SSIRST bit. Because the software reset by the SSIRST bit has priority over the reset by this bit, this bit cannot be set to 1 when the SSIRST bit is 1.

TFRST Bit (Transmit FIFO Reset)

This bit is used to reset the transmit FIFO. Writing 1 to this bit resets the transmit FIFO and initializes the TDC[5:0] bits and TDE flag in the SSIFSR register. Since this bit is not automatically cleared, write 0 to this bit to release from the reset state. After writing 0 to this bit, be sure to check that this bit is 0 before starting the next procedure.

This bit is subject to the software reset by the SSIRST bit. Because the software reset by the SSIRST bit has priority over the reset by this bit, this bit cannot be set to 1 when the SSIRST bit is set.

RIE Bit (Receive Data Full Interrupt Enable)

This bit enables or disables a receive data full interrupt. Set this bit to 1 after specifying the condition for generating a receive data full interrupt in the SSISCR.RDFS[4:0] bits.

TIE Bit (Transmit Data Empty Interrupt Enable)

This bit enables or disables a transmit data empty interrupt. Set this bit to 1 after specifying the condition for generating a transmit data empty interrupt in the SSISCR.TDES[4:0] bits.

BSW Bit (Byte Swap)

The BSW bit enables or disables byte swapping when the SSIFTDR is written and the SSIFRDR register is read. This bit is valid only when the SSIFTDR and SSIFRDR register are accessed in 16-bit or 32-bit units. For details, refer to Figure 47.6.

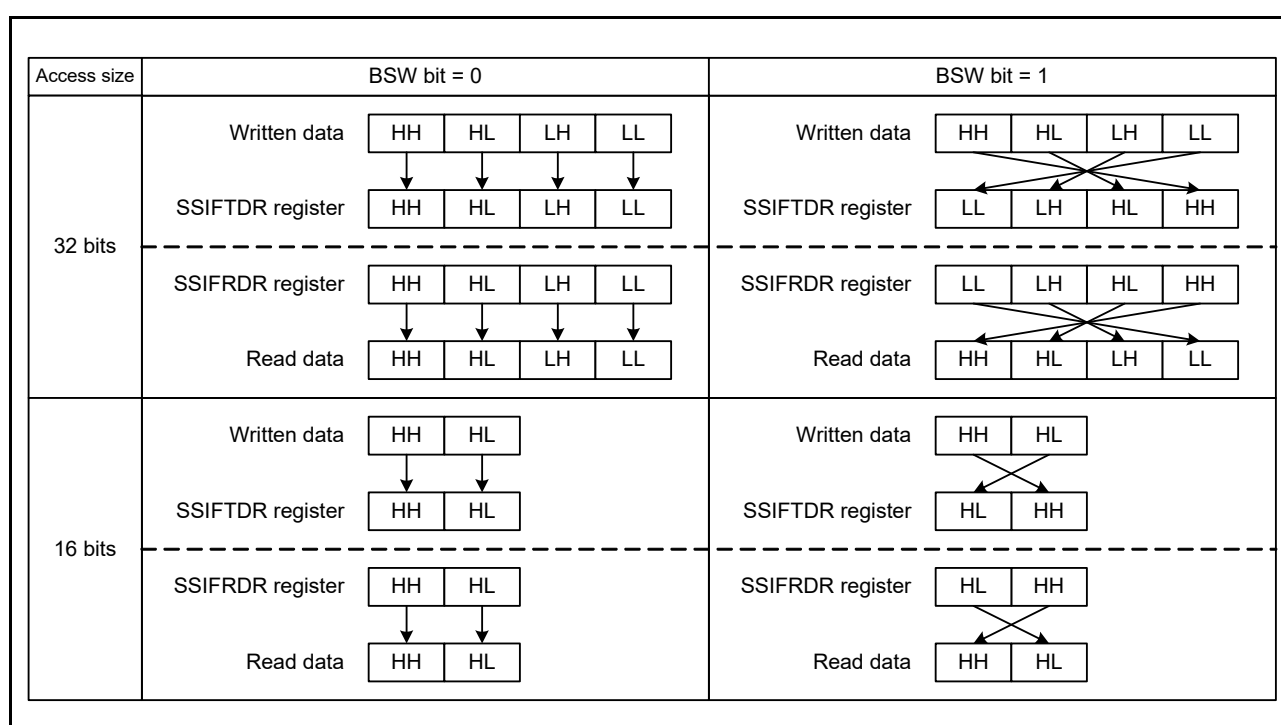


Figure 47.6 Examples of Byte-Swapping Operations

SSIRST Bit (Software Reset)

This bit is used to reset SSIE. Writing 1 to this bit initializes the internal state of SSIE. The registers and bits to be reset by setting this bit are listed below.

- SSICR.MUEN, TEN, and REN bits
- SSISR register
- SSIFCR.TFRST and RFRST bits
- SSIFSR register
- SSIFTDR register
- SSIFRDR register

Because this bit is not automatically cleared, write 0 to this bit to release from the reset state. After writing 0 to this bit, be sure to check that this bit is 0 before starting the next procedure.

Because writing 1 to this bit during data transfer leads to initialization by software reset regardless of the bit clock, be careful to stop data transfer by using this bit.

AUCKE Bit (MCK Supply Enable)

This bit enables/disables supply of MCK while in master-mode. The setting value of this bit is reflected at the timing shown in Figure 47.7.

Changes to settings related to MCK (the settings of the MST, BCKP, and CKDV[3:0] bits in the SSICR register) should only proceed while this bit is 0.

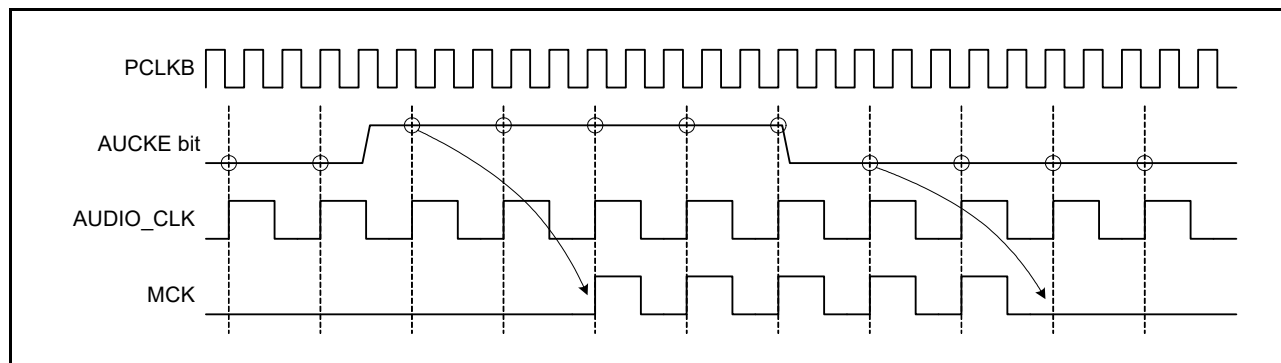


Figure 47.7 Supplying and Stopping MCK

47.2.4 FIFO Status Register (SSIFSR)

Address(es): SSIE0.SSIFSR 0008 A514h, SSIE1.SSIFSR 0008 A554h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	TDC[5:0]					—	—	—	—	—	—	—	—	TDE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	RDC[5:0]					—	—	—	—	—	—	—	—	RDF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	RDF	Receive Data Full Flag	0: The number of received data stored in the receive FIFO does not exceed the value of the SSISCR.RDFS[4:0] bits. 1: The number of received data stored in the receive FIFO exceeds the value of the RDFS[4:0] bits.	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b13 to b8	RDC[5:0]	Receive FIFO Data Count	Number of data stored in the receive FIFO	R
b15, b14	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b16	TDE	Transmit Data Empty Flag	0: The number of spaces available in the transmit FIFO does not exceed the value of the SSISCR.TDES[4:0] bits 1: The number of spaces available in the transmit FIFO exceeds the value of TDES[4:0] bits.	R/W
b23 to b17	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b29 to b24	TDC[5:0]	Transmit FIFO Data Count	Number of data stored in the transmit FIFO	R
b31, b30	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

This register indicates the status of the transmit FIFO and the receive FIFO.

RDF Flag (Receive Data Full Flag)

This flag indicates that the number of unread data in the receive FIFO is more than the value of the SSISCR.RDFS[4:0] bits.

[Priority order for setting and clearing]

Clearing is prioritized.

[Setting condition]

- When the number of data stored in the receive FIFO exceeds the value of the SSISCR.RDFS[4:0] bits.

[Clearing condition]*1

- When 0 is written to this flag after confirming this flag is 1.
- When a received data is read from the SSIFRDR register by DMA/DTC transfer (the last transfer of the block if using block transfer).

Note 1. This flag is cleared by a software reset and receive FIFO reset.

RDC[5:0] Bits (Receive FIFO Data Count)

These bits indicate the number of valid data that are stored in the receive FIFO. When this bit is 00h, there is no received data. When 20h, the receive FIFO is full and there is no space available.

TDE Flag (Transmit Data Empty Flag)

This flag indicates that the number of spaces available in the transmit FIFO is more than the value of the SSISCR.TDES[4:0] bits.

[Priority order for setting and clearing]

Clearing is prioritized.

[Setting condition]

- When the number of spaces available in the transmit FIFO exceeds the value of the SSISCR.TDES[4:0] bits.

[Clearing condition]*1

- When 0 is written to this flag after confirming this flag is 1.
- When a transmit data is written to the SSIFTDR register by DMA/DTC transfer (the last transfer of the block if using block transfer).

Note 1. This flag is cleared by a software reset and transmit FIFO reset.

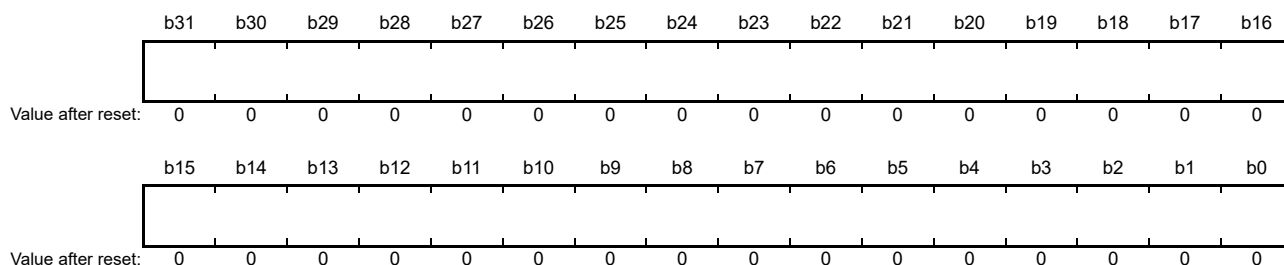
TDC[5:0] Bits (Transmit FIFO Data Count)

These bits indicate the number of valid data that are stored in the transmit FIFO. When this bit is 00h, there is no data to be transmitted. When 20h, the transmit FIFO is full and there is no space available.

47.2.5 Transmit FIFO Data Register (SSIFTDR)

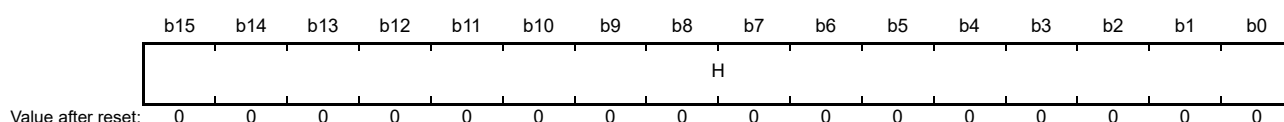
- When accessing in longword size

Address(es): SSIE0.SSIFTDR 0008 A518h, SSIE1.SSIFTDR 0008 A558h



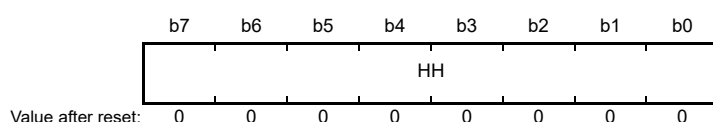
- When accessing in word size

Address(es): SSIE0.SSIFTDR.H 0008 A518h, SSIE1.SSIFTDR.H 0008 A558h



- When accessing in byte size

Address(es): SSIE0.SSIFTDR.HH 0008 A518h, SSIE1.SSIFTDR.HH 0008 A558h



This register is a write-only register used to write a transmit data to the 32-stage and 4-byte-wide transmit FIFO. Access to this register according to the access size listed in Table 47.6.

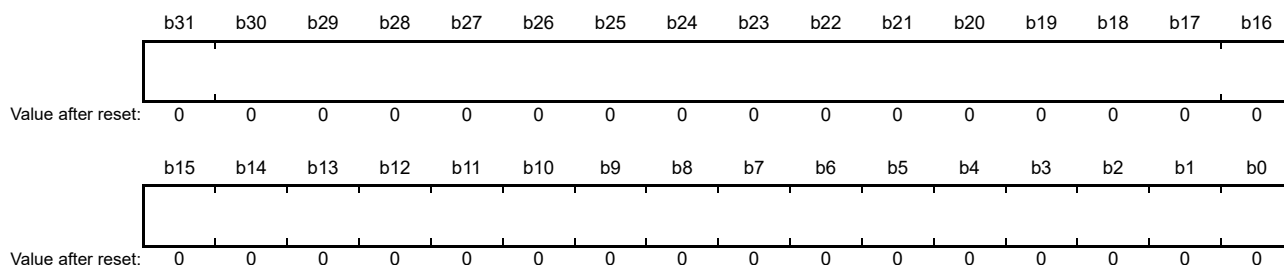
Table 47.6 Access Size to the SSIFTDR and SSIFRDR Registers

SSICR.DWL[2:0] Bits	Data Word Length	Access Size		
		Byte	Word	Longword
000b	8	Available	Not available	Not available
001b	16	Not available	Available	Not available
010b	18	Not available	Not available	Available
011b	20	Not available	Not available	Available
100b	22	Not available	Not available	Available
101b	24	Not available	Not available	Available
110b	32	Not available	Not available	Available

47.2.6 Receive FIFO Data Register (SSIFRDR)

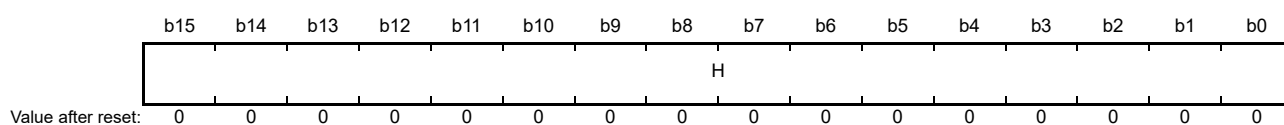
- When accessing in longword size

Address(es): SSIE0.SSIFRDR 0008 A51Ch, SSIE1.SSIFRDR 0008 A55Ch



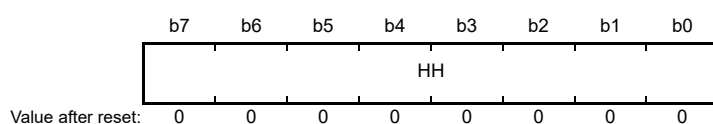
- When accessing in word size

Address(es): SSIE0.SSIFRDR.H 0008 A51Ch, SSIE1.SSIFRDR.H 0008 A55Ch



- When accessing in byte size

Address(es): SSIE0.SSIFRDR.HH 0008 A51Ch, SSIE1.SSIFRDR.HH 0008 A55Ch



This register is a read-only register used to read a received data from the 32-stage and 4-byte-wide receive FIFO. Access to this register according to the access size listed in Table 47.6.

47.2.7 Audio Format Register (SSIOFR)

Address(es): SSIE0.SSIOFR 0008 A520h, SSIE1.SSIOFR 0008 A560h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	BCKASTP	LRCONT	—	—	—	—	—	—	OMOD[1:0]	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b1, b0	OMOD[1:0]	Data Format Select*1	b1 b0 0 0: I ² S, left-justified, or right-justified format 0 1: TDM format 1 0: Monaural format 1 1: Setting prohibited	R/W
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b8	LRCONT	LRCK Continuous Output Enable*2, *3	0: LRCK stops in the idle state. 1: Output of LRCK continues in the idle state.	R/W
b9	BCKASTP	BCK Continuous Output Disable*2, *3	0: BCK is always output. 1: BCK stops in idle state.	R/W
b31 to b10	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Rewrite these bits while SSIE is in idle state (SSISR.IIRQ flag = 1). If the value of these bits is changed while data transfer is in progress, the operation after the change is not guaranteed.

Note 2. This bit is valid only in master-mode. The setting is invalid in slave-mode.

Note 3. Do not set both of the BCKASTP and LRCONT bits to 1.

This register is used to set an audio format (which includes the settings of data format, LR clock continuous output mode, and BCK continuous output disable).

LRCONT Bit (LRCK Continuous Output Enable)

This bit enables or disables the output from SSILRCK pin when SSIE is in the idle state. This bit is effective only in master mode.

BCKASTP Bit (BCK Continuous Output Disable)

This bit enables or disables the output from SSIBCK pin when SSIE is in the idle state. This bit is effective only in master mode.

Set this bit to 0 to output BCK from the SSIBCK pin before starting data transfer. If this bit is set to 1 during data transfer, BCK output from the SSIBCK pin stops when the SSIE enters the idle state.

Set this bit after the data format to be used is specified. Do not rewrite this bit while the supply of MCK stops.

47.2.8 FIFO Status Control Register (SSISCR)

Address(es): SSIE0.SSISCR 0008 A524h, SSIE1.SSISCR 0008 A564h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	TDES[4:0]				—	—	—	RDFS[4:0]				—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b4 to b0	RDFS[4:0]	Receive Data Full Condition Setting*1	Specifies the condition for setting the RDF flag to 1. b4 b0 00000: When receive FIFO has unread received data. 00001: When receive FIFO has two or more unread received data. ... 11110: When receive FIFO has 31 or more unread received data. 11111: When receive FIFO is full.	R/W
b7 to b5	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b12 to b8	TDES[4:0]	Transmit Data Empty Condition Setting*1	Specifies the condition for setting the TDE flag to 1. b12 b8 00000: When transmit FIFO has space available. 00001: When transmit FIFO has at least two stages available. ... 11110: When transmit FIFO has at least 31 stages available. 11111: When transmit FIFO is empty.	R/W
b31 to b13	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Rewrite these bits while SSIE is in idle state (SSISR.IIRQ flag = 1). If the value of these bits is changed while data transfer is in progress, the operation after the change is not guaranteed.

47.3 Operation

47.3.1 Data Formats

SSIE supports five data formats. Table 47.7 lists the supported data formats.

Table 47.7 Supported Data Formats and Bit Setting

Data Format	SSIOFR.OMOD[1:0] Bits	SSICR.DEL Bit	SSICR.LRCKP Bit	SSICR.SDTA Bit
I ² S format	00b	0	0	0
Left-justified format	00b	1	1	0
Right-justified format	00b	1	1	1
TDM format	01b	0 or 1	1	0 or 1
Monaural format	10b	0 or 1	1	0 or 1

The following describes the serial data structure shared by data formats.

A serial data structure is defined by the system word length (set in the SSICR.SWL[2:0] bits) and the data word length (set in the SSICR.DWL[2:0] bits). If the data word length is shorter than the system word length, padding bits are inserted (Figure 47.8).

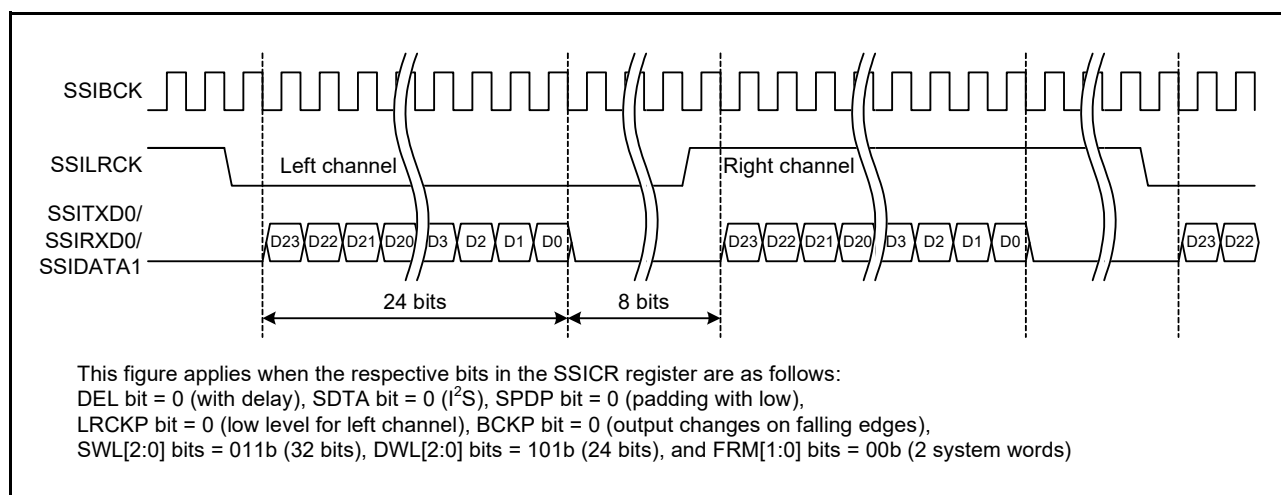


Figure 47.8 Example of Padding Bit Insertion (I²S Format, System Word Length > Data Word Length)

The number of padding bits is calculated from the system word length minus the data word length.

Table 47.8 lists the number of padding bits to be inserted with each combination of system word length (SWL[2:0] bits) and data word length (DWL[2:0] bits).

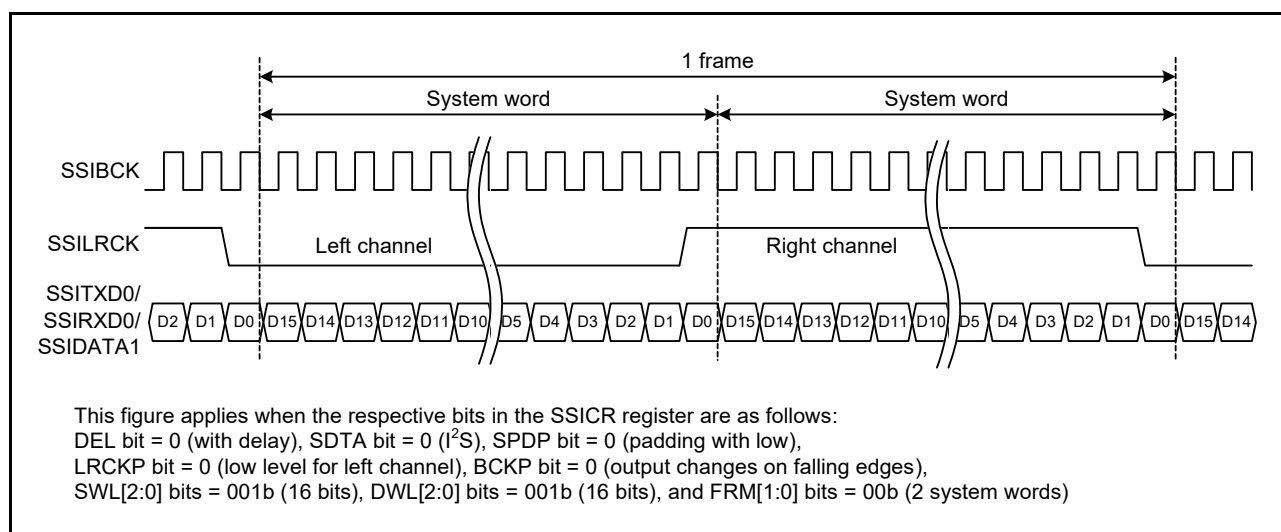
Table 47.8 Number of Padding Bits

System Word Length (SSICR.SWL[2:0] Bits)	Data Word Length (SSICR.DWL[2:0] Bits)						
	8 (000b)	16 (001b)	18 (010b)	20 (011b)	22 (100b)	24 (101b)	32 (110b)
8 (000b)	0	Prohibited setting	Prohibited setting	Prohibited setting	Prohibited setting	Prohibited setting	Prohibited setting
16 (001b)	8	0	Prohibited setting	Prohibited setting	Prohibited setting	Prohibited setting	Prohibited setting
24 (010b)	16	8	6	4	2	0	Prohibited setting
32 (011b)	24	16	14	12	10	8	0
48 (100b)	40	32	30	28	26	24	16
64 (101b)	56	48	46	44	42	40	32
128 (110b)	120	112	110	108	106	104	96
256 (111b)	248	240	238	236	234	232	224

47.3.1.1 I²S Format

The I²S format configures a frame with two system words of left channel and right channel. The SSILRCK signals becomes low for left channel and high for right channel. The effective data are output one bit clock period after the SSILRCK changes and then the padding bits are output.

Figure 47.9 shows an example of I²S format.

**Figure 47.9** Example of I²S Format

47.3.1.2 Left-Justified Format

The left-justified format configures a frame with two system words of left channel and right channel. The SSILRCK signals becomes high for left channel and low for right channel. The effective data are output at the same time as the SSILRCK changes and then the padding bits are output.

Figure 47.10 shows an example of left-justified format.

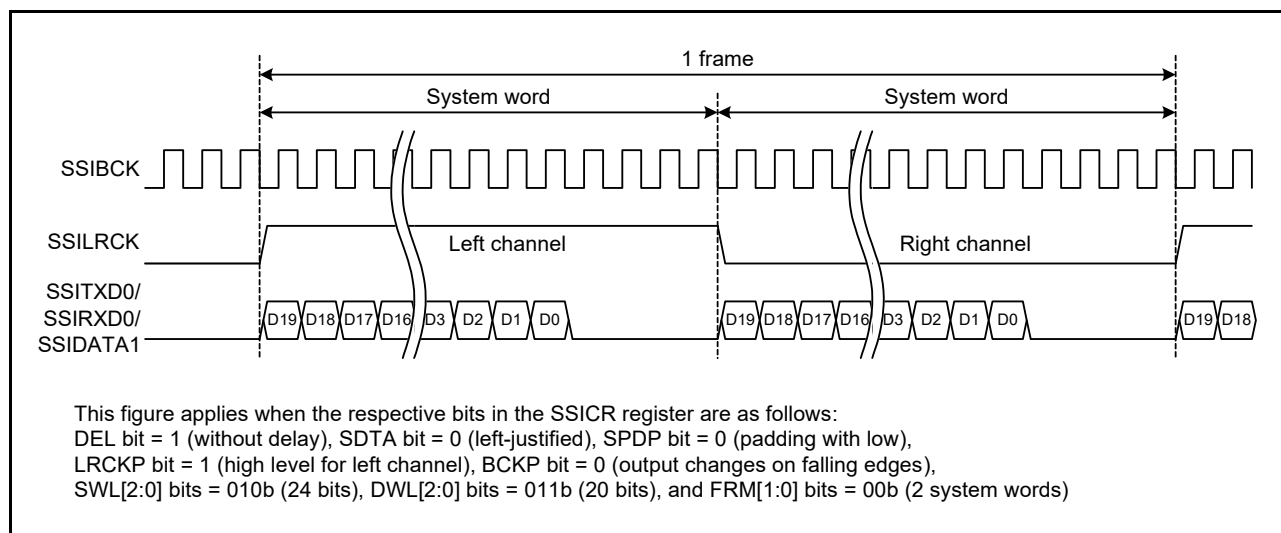


Figure 47.10 Example of Left-Justified Format

47.3.1.3 Right-Justified Format

The right-justified format configures a frame with two system words of left channel and right channel. The SSILRCK signals becomes high for left channel and low for right channel. The padding bits are output at the same time as the SSILRCK changes and then the effective data are output.

Figure 47.11 shows an example of right-justified format.

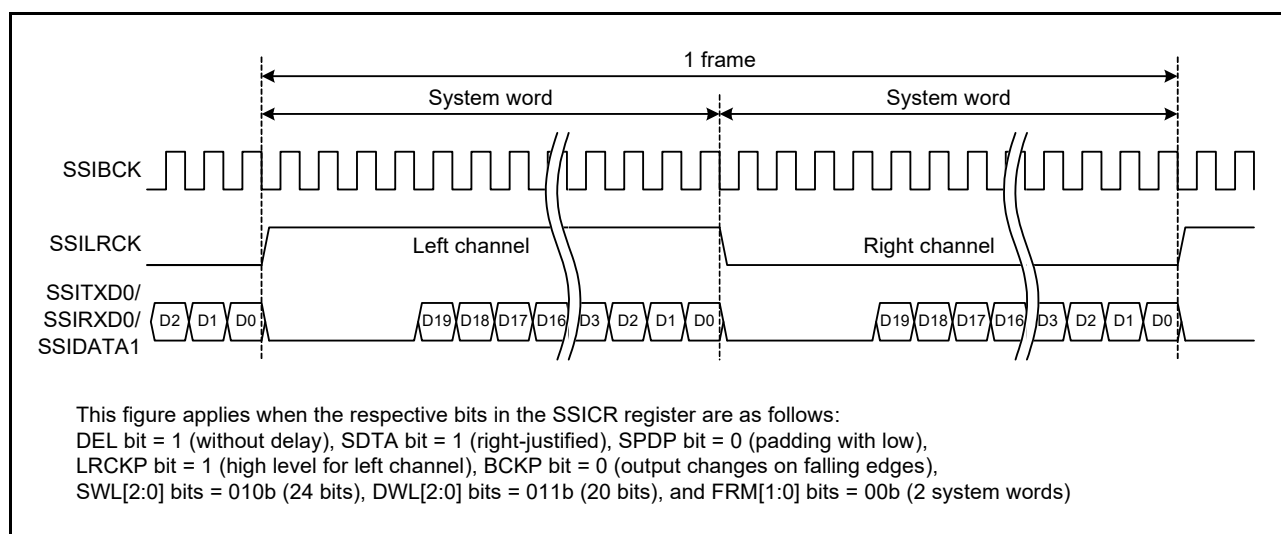


Figure 47.11 Example of Right-Justified Format

47.3.1.4 Monaural Format

The monaural format configures a frame with one system word. The SSILRCK signal is used as a transfer start trigger. Figure 47.12, Figure 47.13, and Figure 47.14 show examples of monaural formats with 0 as the value of the SSICR.DEL bit and no padding bits, with 0 as the value of the DEL bit and padding bits, and with 1 as the value of the DEL bit and no padding bits, respectively.

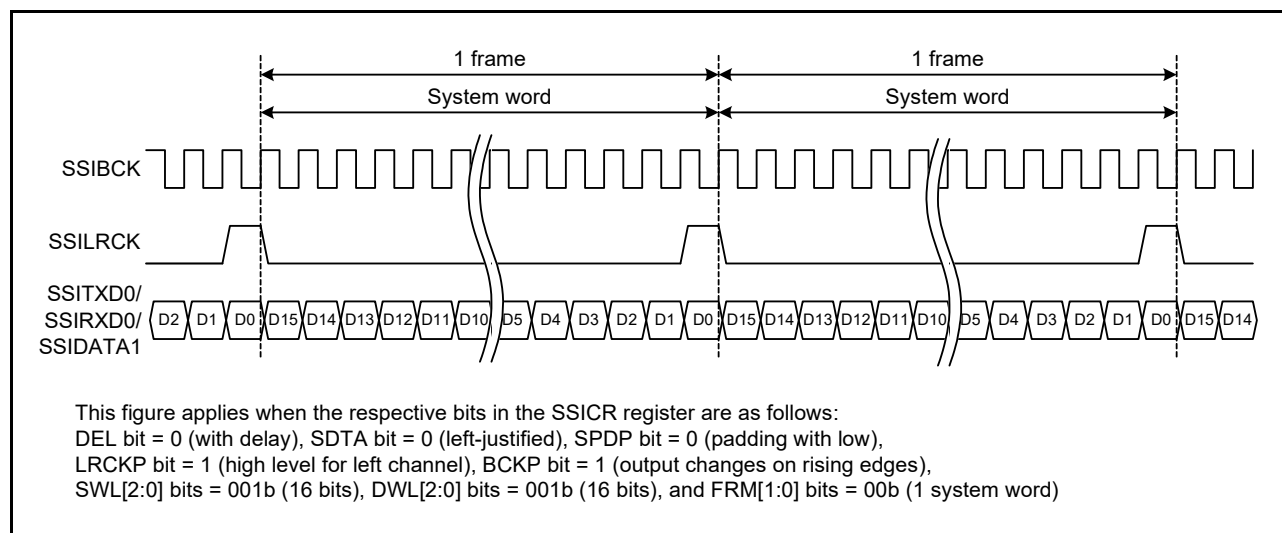


Figure 47.12 Example of Monaural Format (1): SSICR.DEL Bit = 0 and No Padding

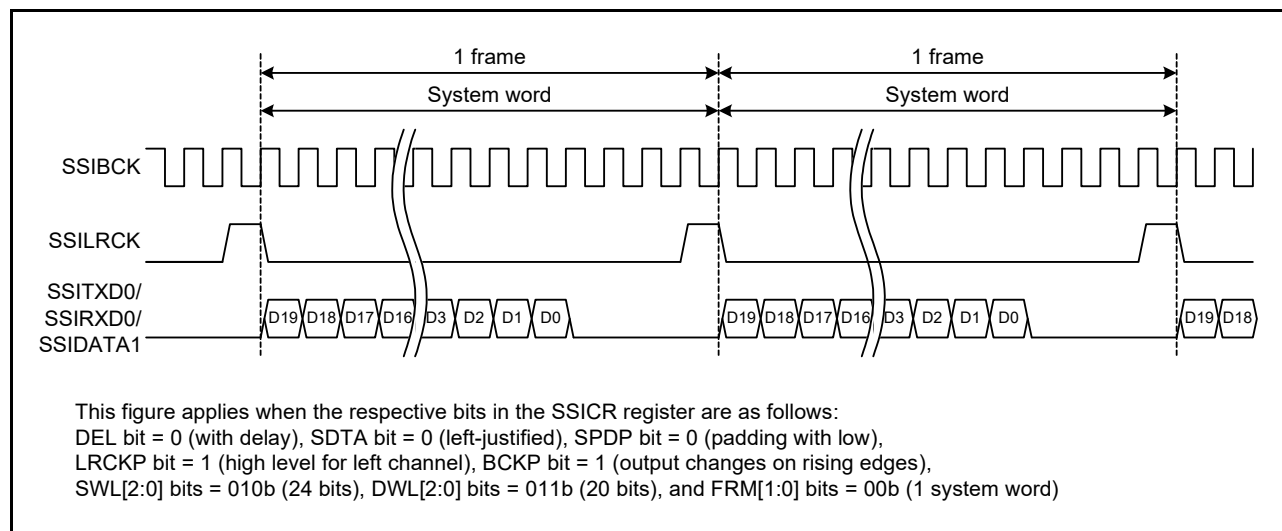


Figure 47.13 Example of Monaural Format (2): SSICR.DEL Bit = 0 and Padding Bits

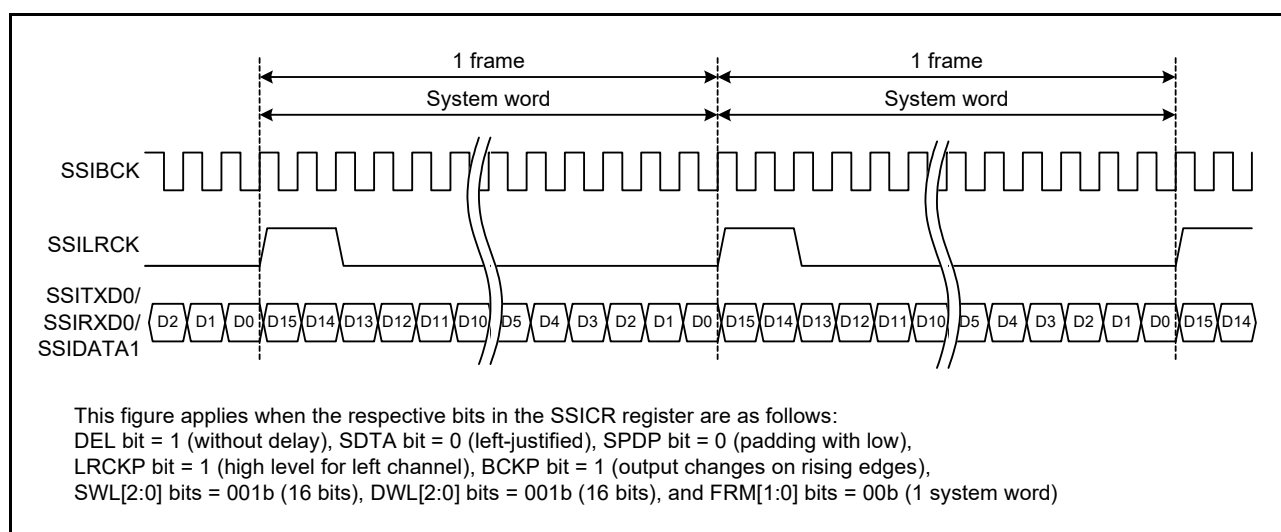


Figure 47.14 Example of Monaural Format (3): SSICR.DEL Bit = 1 and No Padding

47.3.1.5 TDM Format

The TDM format configures a frame with four to eight system words. The SSILRCK signals become high for the first system word and low for the other system words. The SSILRCK signal is defined as the synchronous pulse and its rising edge means a start of a frame.

Figure 47.15 and Figure 47.16 show examples of the TDM formats with 0 as the value of the SSICR.DEL bit and no padding bits, and with 1 as the value of the SSICR.DEL bit and padding bits, respectively.

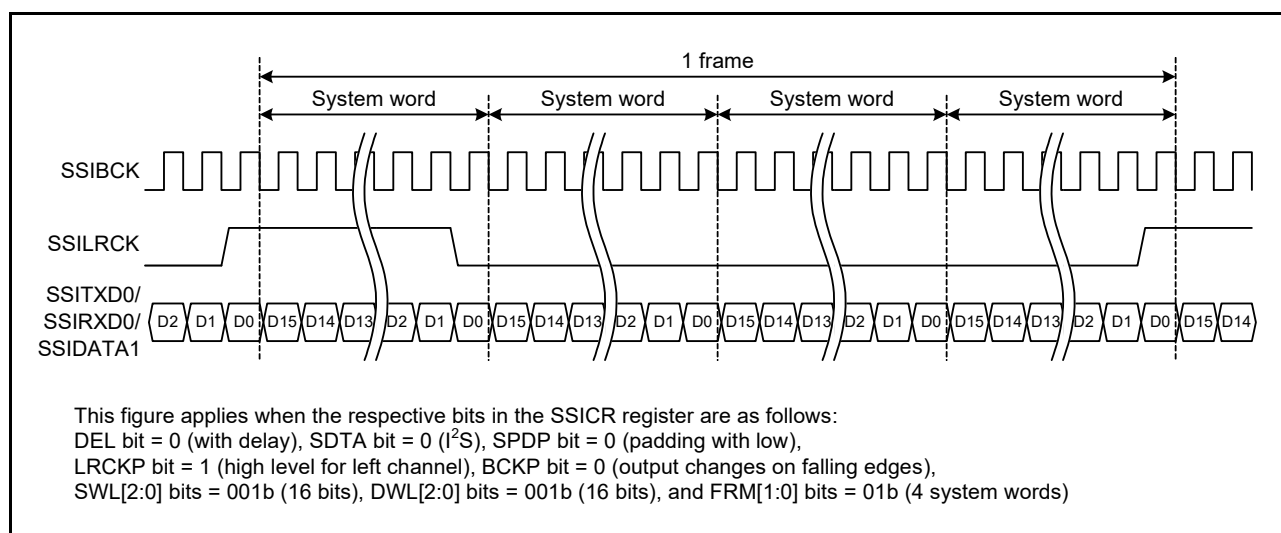


Figure 47.15 Example of TDM Format (1): SSICR.DEL Bit = 0 and No Padding

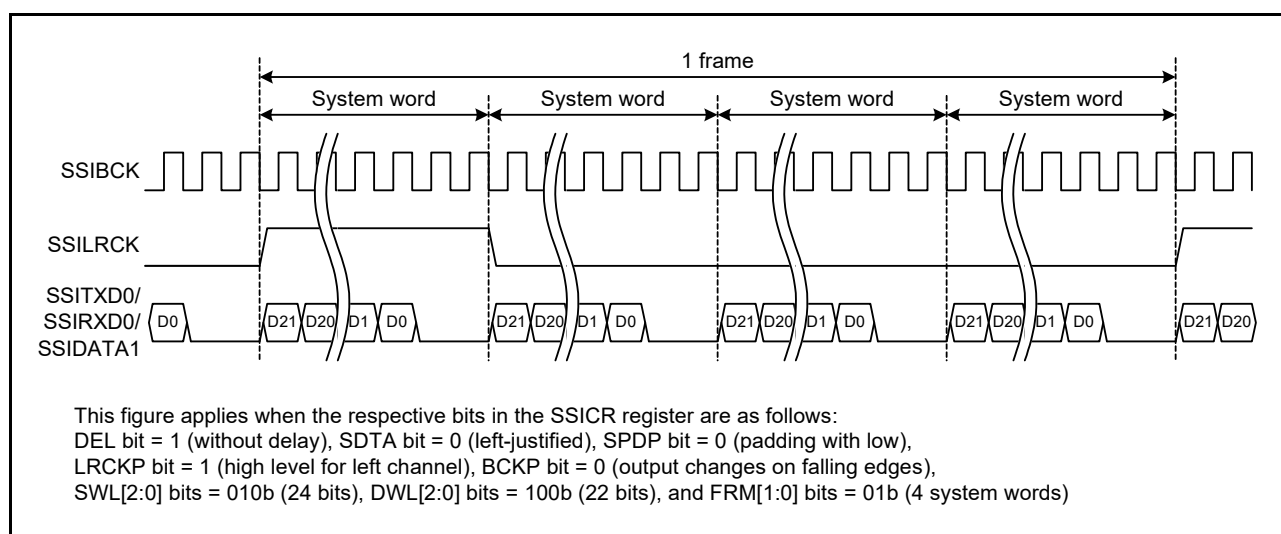


Figure 47.16 Example of TDM Format (2): SSICR.DEL Bit = 1 and Padding Bits

47.3.2 Output Control

SSIE has functions to control the outputs of data and clocks such as mute function, LRCK continuous output function, and BCK continuous output disable function.

47.3.2.1 Mute Function

The output from the SSITXD0 or SSIDATA1 pin can be fixed to the low level by using the mute function. If the SSICR.MUEN bit is changed during a frame, the setting takes effect at the next frame boundary. Even while the MUEN bit is set to 1, status flags are updated and interrupt request signals are generated.

Figure 47.17 shows an example of the mute operation.

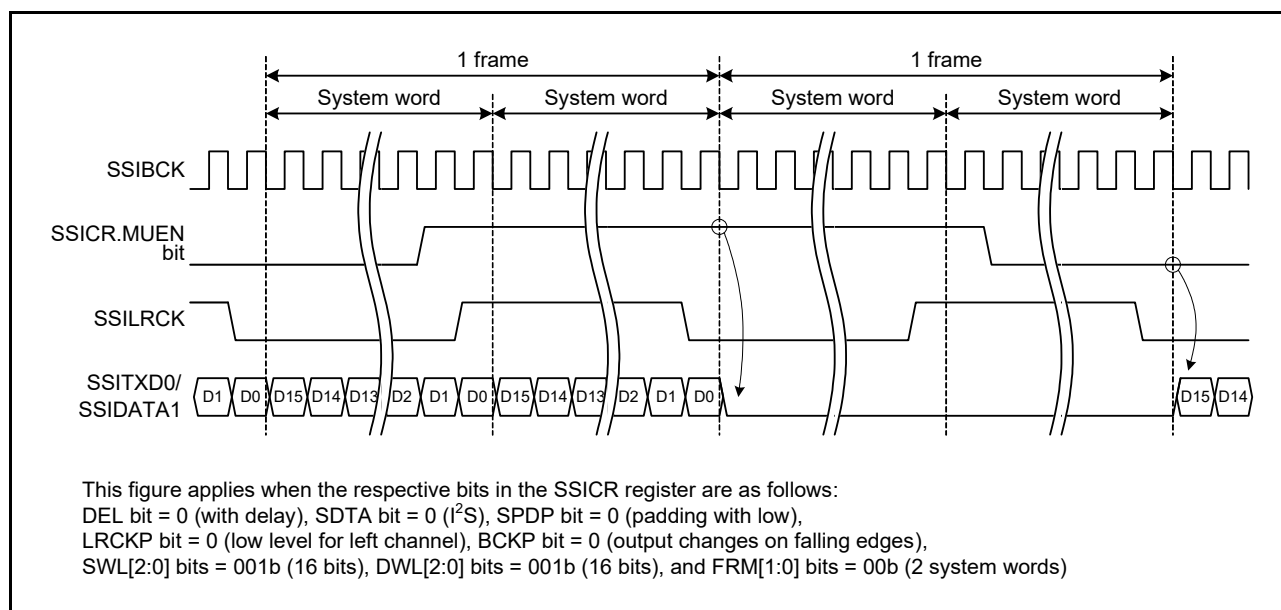


Figure 47.17 Example of Mute Operation

47.3.2.2 LRCK Continuous Output Function

LRCK can be output from the SSILRCK pin even in the idle state by using the LRCK continuous output function. It is controlled by the SSIOFR.LRCONT bit.

Figure 47.18 shows an example of the operation with LRCK continuous output function.

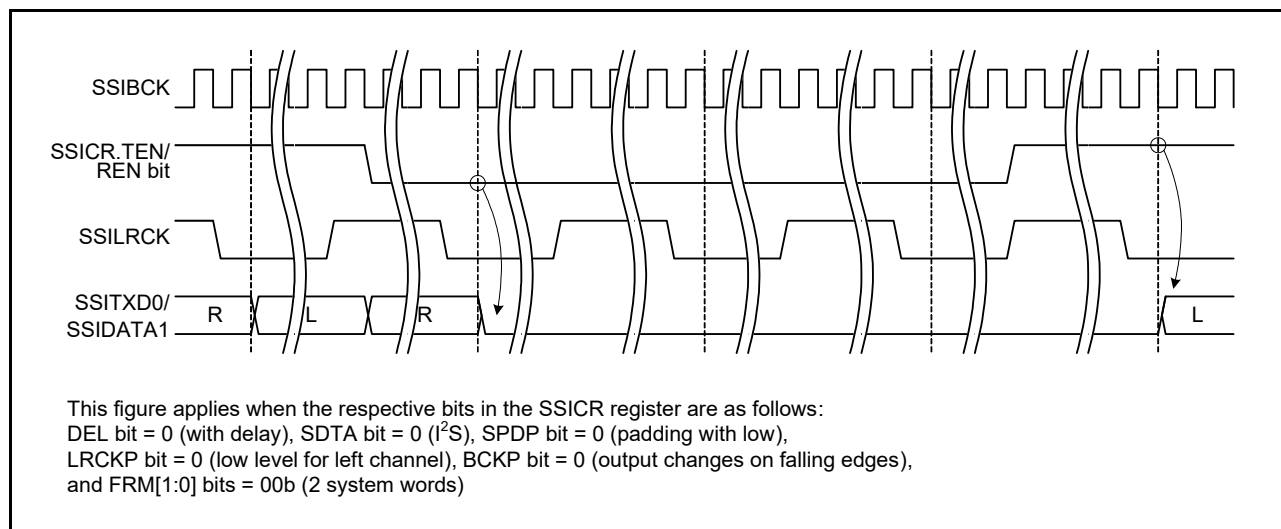


Figure 47.18 Example of Operation with LRCK Continuous Output Function

47.3.2.3 BCK Continuous Output Disable Function

BCK output from the SSIBCK pin can be stopped in the idle state by using the BCK continuous output disable function. It is controlled by the SSIOFR.BCKASTP bit. When the BCKASTP bit is set to 1, BCK stops in the idle state.

Set the BCKASTP bit to 0 and then start data transfer. If this bit is set to 1 during data transfer, BCK stops when SSIE enters the idle state. To resume the data transfer, set the BCKSTP bit to 0 to output BCK and then set the SSICR.TEN/REN bit to 1.

Figure 47.19 shows an example of the operation with BCK continuous output disable function.

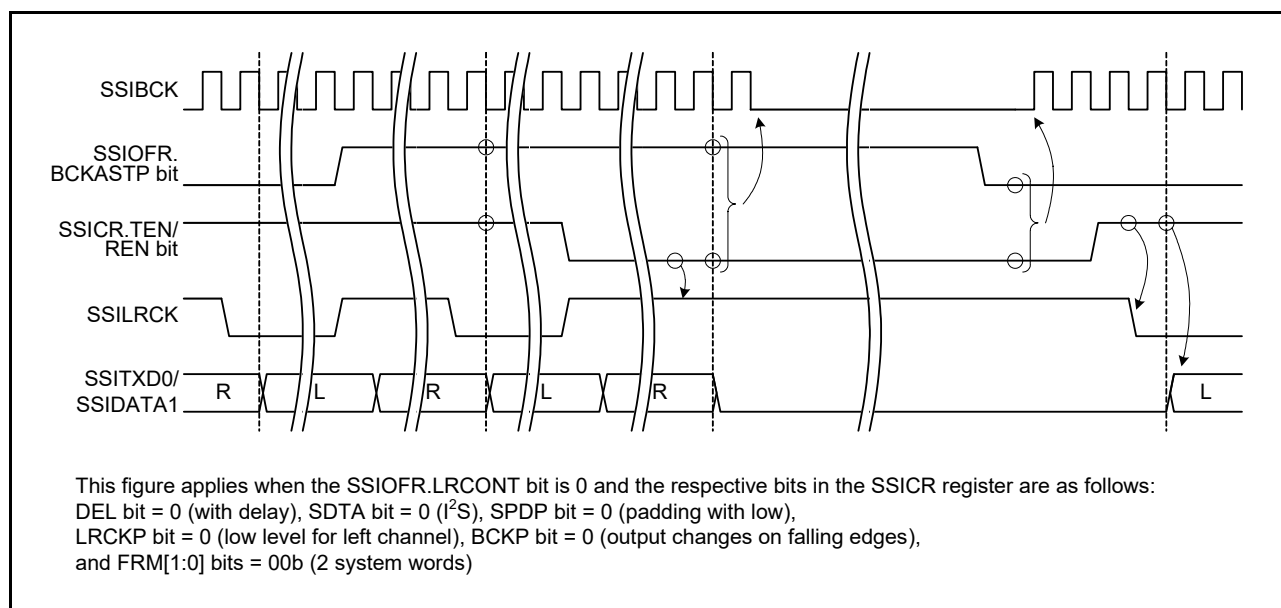


Figure 47.19 Example of Operation with BCK Continuous Output Disable Function

47.3.3 Transfer Modes

SSIE supports the transfer modes listed in Table 47.9. Table 47.10 lists the control bits valid in each transfer mode.

Table 47.9 Transfer Modes

Transfer Mode	SSICR.MST Bit	SSICR.REN Bit	SSICR.TEN Bit
Slave-mode transmission	0	0	1
Slave-mode reception	0	1	0
Slave-mode transception	0	1	1
Master-mode transmission	1	0	1
Master-mode reception	1	1	0
Master-mode transception	1	1	1

Table 47.10 Control Bits Valid in Each Transfer Mode*1

Control Bit		Transfer Mode					
Register	Bit	Slave-mode Reception	Slave-mode Transmission	Slave-mode Transception	Master-mode Reception	Master-mode Transmission	Master-mode Transception
SSICR	CKDV[3:0]	Ineffective	Ineffective	Ineffective	Effective	Effective	Effective
	MUEN	Ineffective	Effective	Effective	Ineffective	Effective	Effective
SSIFCR	AUCKEN	Ineffective	Ineffective	Ineffective	Effective	Effective	Effective
	TIE	Ineffective	Effective	Effective	Ineffective	Effective	Effective
	RIE	Effective	Ineffective	Effective	Effective	Ineffective	Effective
	TFRST	Ineffective	Effective	Effective	Ineffective	Effective	Effective
	RFRST	Effective	Ineffective	Effective	Effective	Ineffective	Effective
SSIOFR	BCKASTP	Ineffective	Ineffective	Ineffective	Effective	Effective	Effective
	LRCONT	Ineffective	Ineffective	Ineffective	Effective	Effective	Effective
	OMOD[1:0]	Effective	Effective	Effective	Effective	Effective	Effective
SSISCR	TDES[4:0]	Ineffective	Effective	Effective	Ineffective	Effective	Effective
	RDFS[4:0]	Effective	Ineffective	Effective	Effective	Ineffective	Effective

Note 1. "Ineffective" means that the setting of the bits does not affect operation.

47.3.3.1 Slave Mode

The SSIE operates in slave mode when the SSICR.MST bit is 0. The SSIBCK and SSILRCK signals must be supplied from an external device.

47.3.3.2 Master Mode

The SSIE operates in master mode when the SSICR.MST bit is 1. The SSIBCK and SSILRCK signals are internally generated from the audio clock (AUDIO_CLK).

47.3.3.3 Transmission

The SSIE operates as a transmitter when the SSICR.TEN bit is 1 and the SSICR.REN bit is 0. Receive circuits do not operate.

47.3.3.4 Reception

The SSIE operates as a receiver when the SSICR.TEN bit is 0 and the SSICR.REN bit is 1. Transmit circuits do not

operate.

47.3.3.5 Transception

The SSIE operates as a transceiver when both of the SSICR.TEN and SSICR.REN bits are 1.

47.3.4 State Transition

SSIE has the two main operation states: idle state and transfer state. The SSICR.TEN and SSICR.REN bits switch between these states. The SSISR.IIRQ flag can be used to confirm in which state SSIE is.

Figure 47.20 shows the state transition of the SSIE.

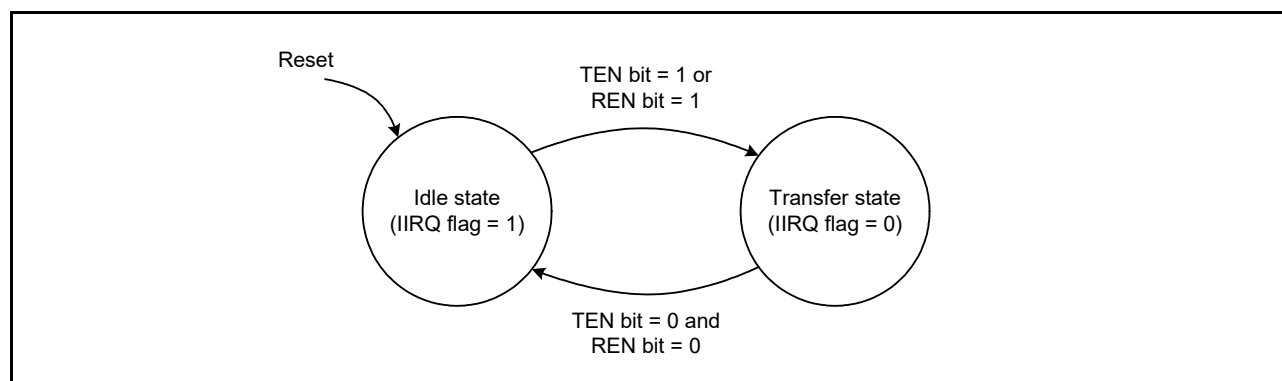


Figure 47.20 State Transition of the SSIE

47.3.4.1 Idle State

In the idle state, SSIE stops data transfer. When both or either of the SSICR.TEN and SSICR.REN bits are/is set to 1, the SSIE enters the transfer state.

Outputs of the BCK and LRCK in master mode can be controlled by the SSIOFR.BCKASTP and SSIOFR.LRCONT bits, respectively. For details, refer to Table 47.11.

Table 47.11 Output of Clock/Data in the Idle State

SSICR.MST Bit	SSIOFR.BCKASTP Bit	SSIOFR.LRCONT Bit	SSIBCK	SSILRCK	SSITXD0/SSIDATA1
0	Invalid	Invalid	— (input)	— (input)	Stop
1	0	0	Supply	Stop	Stop
1	0	1	Supply	Supply	Stop
1	1	0	Stop	Stop	Stop
1	1	1	Stop	Supply	Stop

47.3.4.2 Transfer States

In this state, SSIE continues data transfer. When the SSICR.TEN and SSICR.REN bit is set to 0, SSIE enters the idle state.

For details, refer to Figure 47.21.

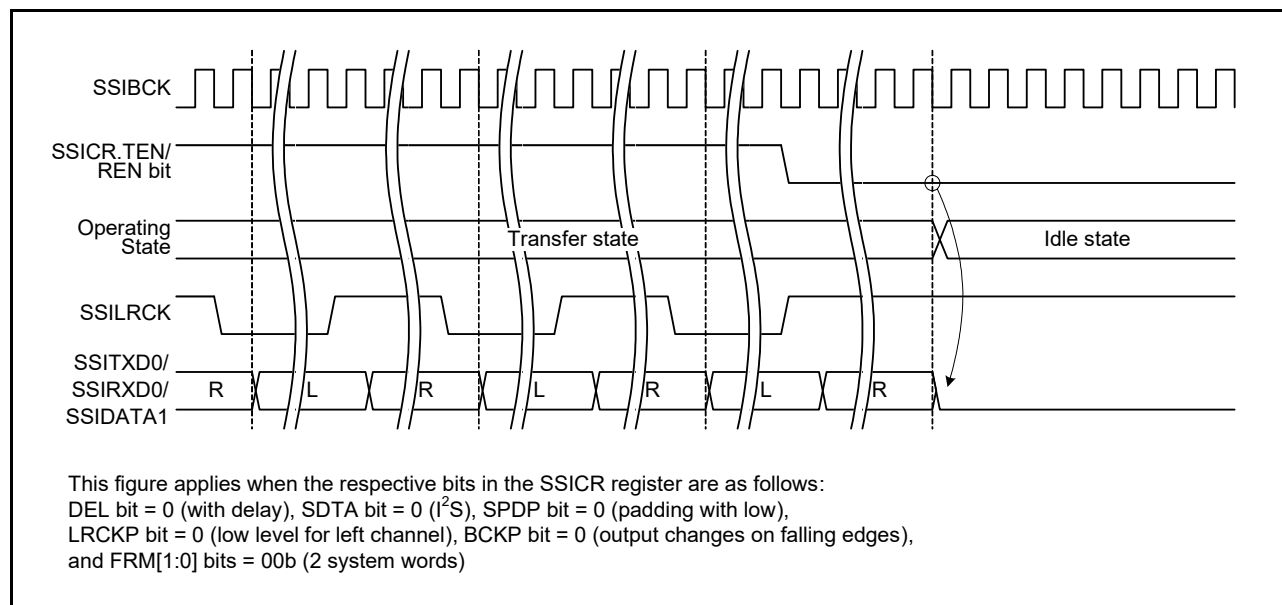


Figure 47.21 Transfer State and Transition to Idle State

47.3.5 SSIE Initialization

Figure 47.22 shows an example of the flow to initialize SSIE.

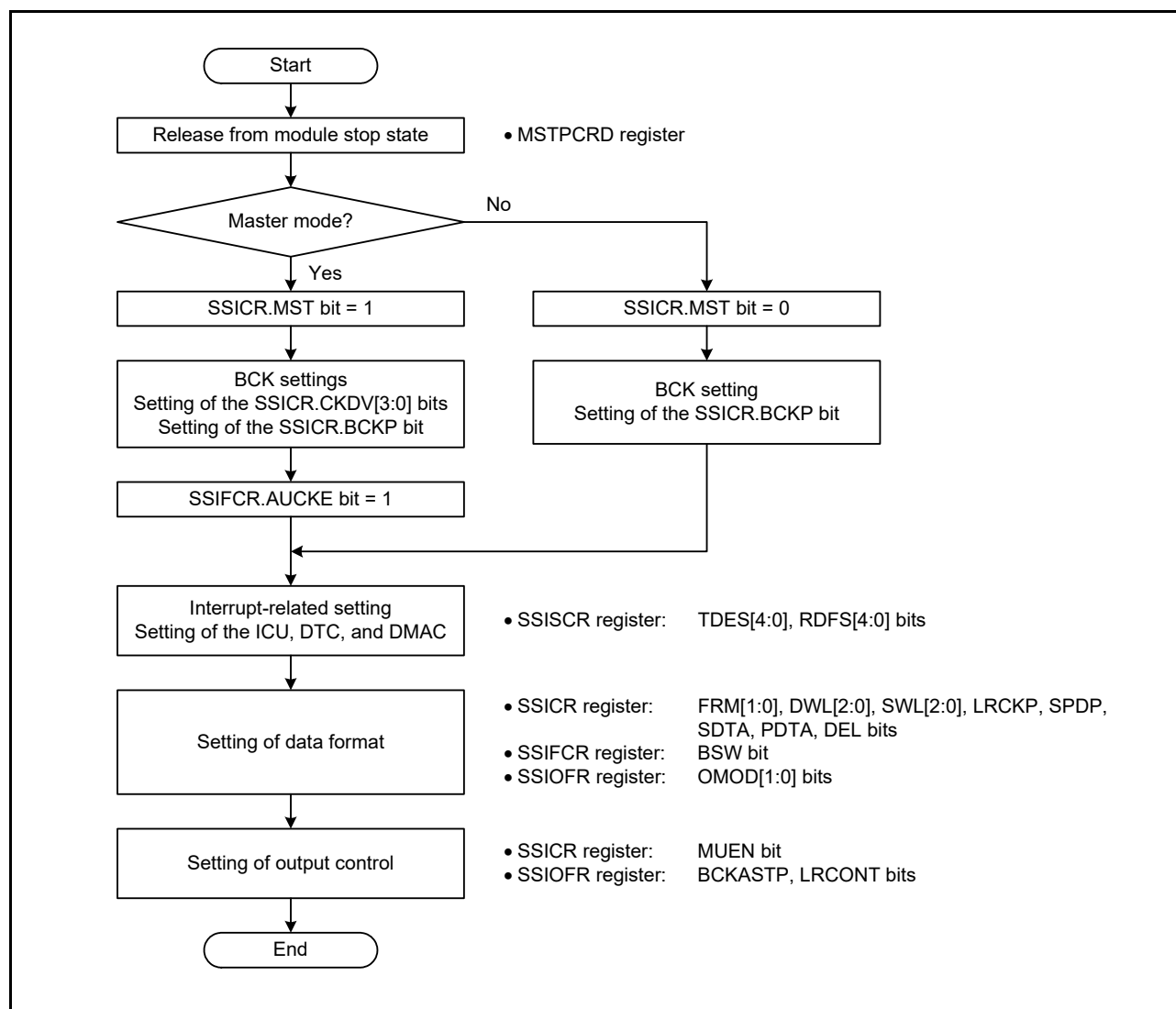


Figure 47.22 Example Flowchart of SSIE Initialization

47.3.6 Transmission

Figure 47.23 shows an example of the flow to transmit data.

When the SSICR.TEN bit is set to 1 after setting the data for one frame in the SSIFTDR register, the SSIE starts transmission. When the number of spaces available in the transmit FIFO reaches the value specified by the SSISCR.TDES[4:0] bits while the SSIFCR.TIE bit is 1, a transmit data empty interrupt occurs. The number of data to be written to the SSIFTDR register should be determined in accordance with the space available in the transmit FIFO specified by the TDES[4:0] bits.

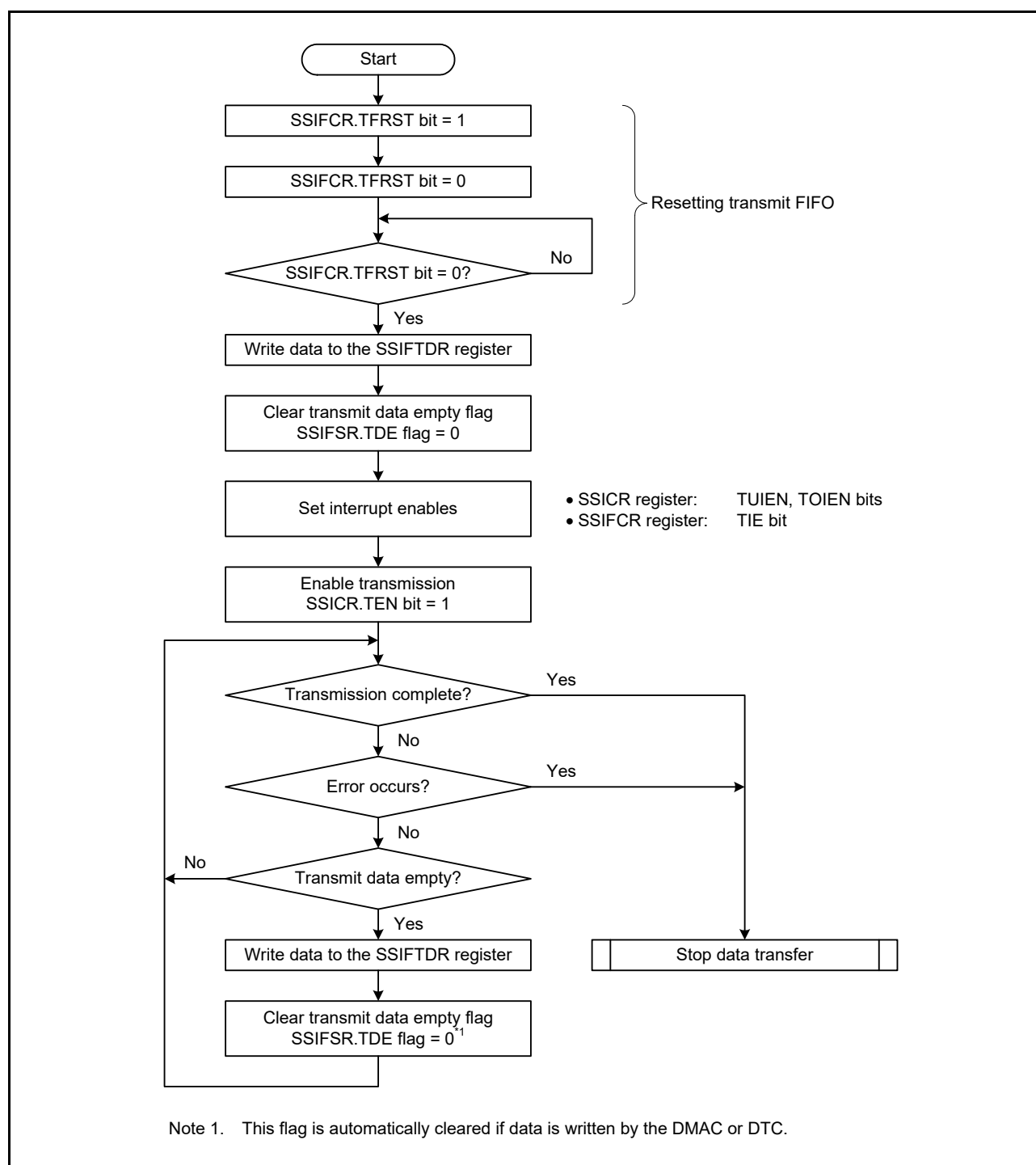


Figure 47.23 Example Flowchart of Transmission

47.3.7 Reception

Figure 47.24 shows an example of the flow to receive data.

When the SSICR.REN bit is set to 1, SSIE starts reception. When the number of data stored in the receive FIFO reaches to the value specified by the SSISCR.RDFS[4:0] bits while the SSIFCR.RIE bit is 1, a receive data full interrupt occurs. The number of data to be read from the SSIFRDR register should be determined in accordance with the number of data stored in the receive FIFO specified by the RDFS[4:0] bits.

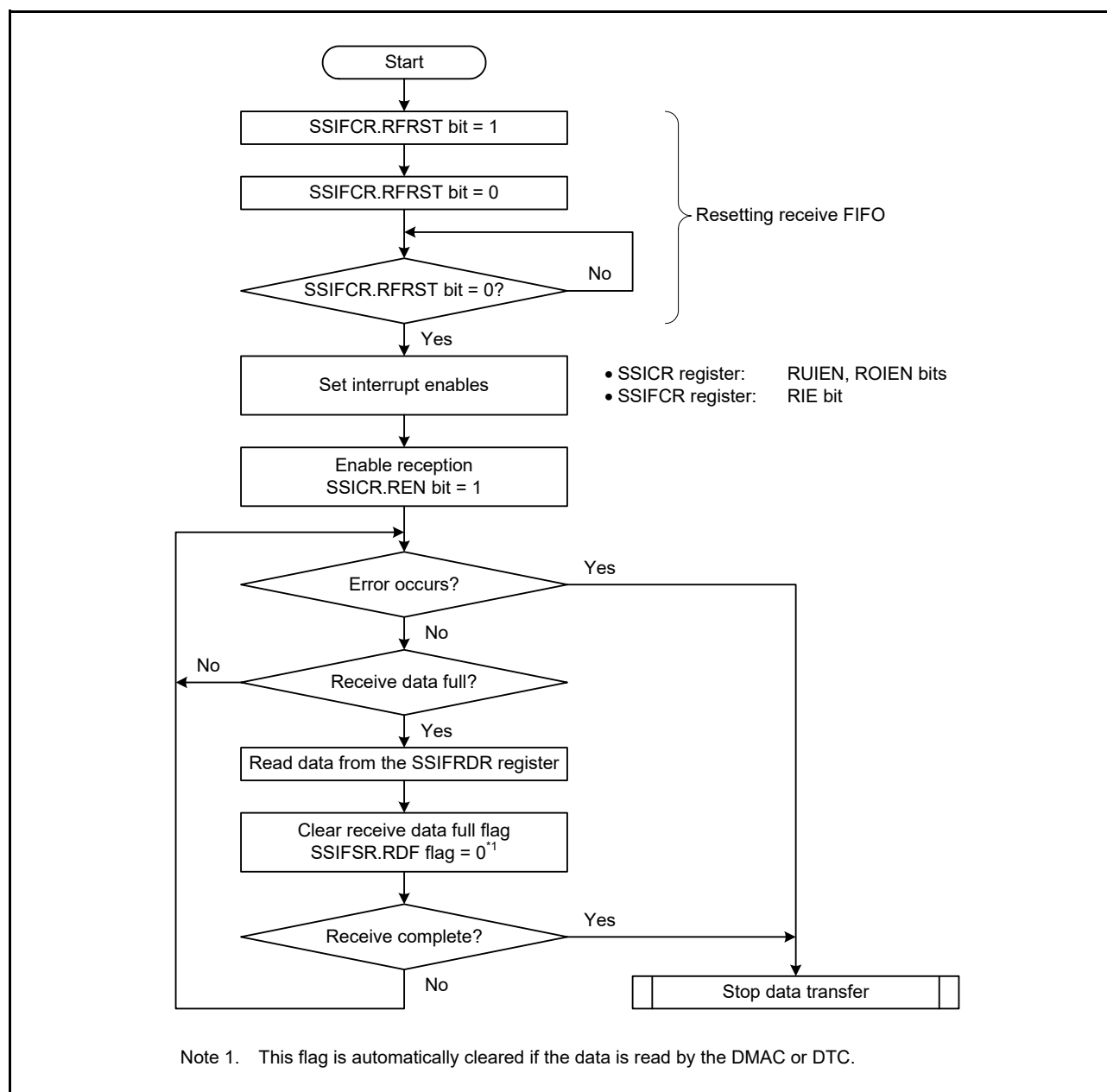


Figure 47.24 Example Flowchart of Reception

47.3.8 Stop Data Transfer

Figure 47.25 shows an example of the flow to stop data transfer.

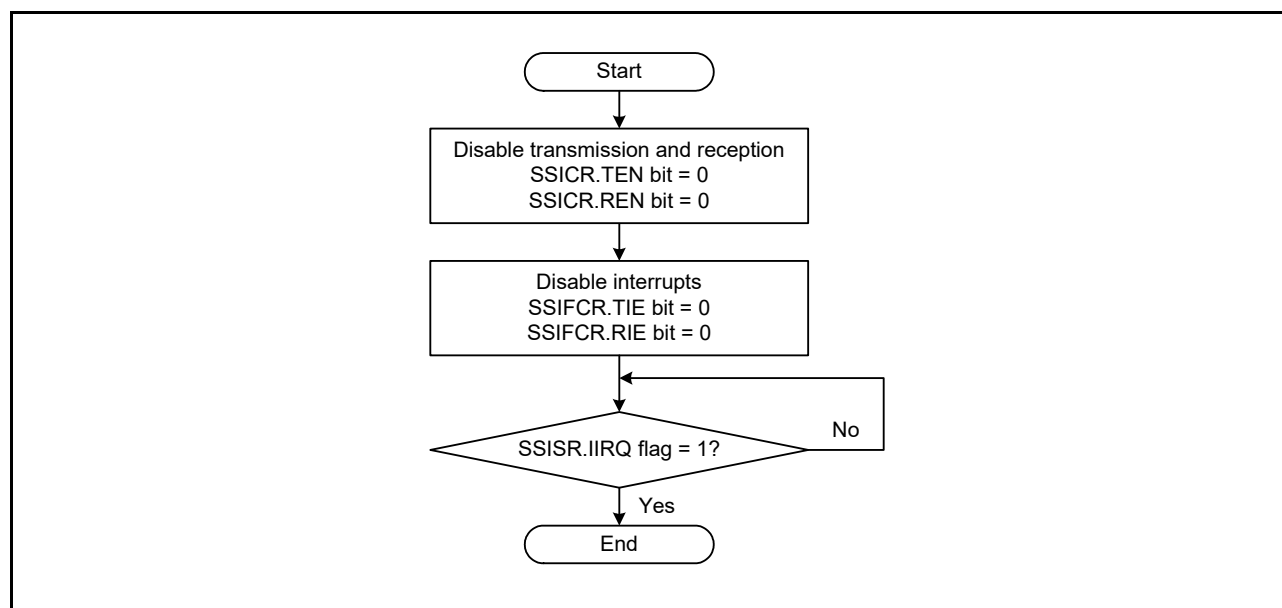


Figure 47.25 Example Flowchart of Stopping Data Transfer

To resume data transfer, conform with the example flowcharts of transmission and reception in Figure 47.23 and Figure 47.24, respectively. When changing the settings of clocks or switching a transfer mode between master and slave, start from initialization.

47.3.9 Error Processing

SSIE has the following four errors.

- Transmit underflow
- Transmit overflow
- Receive underflow
- Receive overflow

If an error has occurred, follow the procedure to stop data transfer shown in Figure 47.25 and then clear the corresponding flag. When an error has occurred while the corresponding interrupt enable bit in the SSICR register is 1, an interrupt is generated.

(1) Transmit Underflow

When a data transfer from the transmit FIFO to transmit shift register has performed while the transmit FIFO is empty, a transmit underflow is informed. If a transmit underflow has occurred, the SSITXD0/SSIDATA1 pin is driven low.

(2) Transmit Overflow

When a data is written to the SSIFTDR register while the transmit FIFO is full, a transmit overflow is informed. The data to be written is discarded.

(3) Receive Underflow

When a data is read from the SSIFRDR register while the receive FIFO is empty, a receive underflow is informed. The read data is undefined.

(4) Receive Overflow

When a data transfer from receive shift register to the receive FIFO has performed while the receive FIFO is full, a receive overflow is informed. The data to be transferred is discarded and is not stored in the receive FIFO.

47.4 Interrupts

Table 47.12 lists the interrupt sources. Figure 47.26 shows an block diagram for interrupts.

Table 47.12 SSIE Interrupt Sources

Channel	Name	Interrupt Source	Interrupt Request Flag	Interrupt Enable Bit	DMAC/DTC Triggerable
SSIE0	SSIF0	Transmit underflow interrupt	SSISR.TUIRQ	SSICR.TUIEN	No
		Transmit overflow interrupt	SSISR.TOIRQ	SSICR.TOIEN	
		Receive underflow interrupt	SSISR.RUIRQ	SSICR.RUIEN	
		Receive overflow interrupt	SSISR.ROIRQ	SSICR.ROIEN	
		Idle state interrupt	SSISR.IIRQ	SSICR.IIEN	
	SSIRX10	Receive data full interrupt	SSIFSR.RDF	SSIFCR.RIE	Yes
	SSITX10	Transmit data empty interrupt	SSIFSR.TDE	SSIFCR.TIE	Yes
SSIE1	SSIF1	Transmit underflow interrupt	SSISR.TUIRQ	SSICR.TUIEN	No
		Transmit overflow interrupt	SSISR.TOIRQ	SSICR.TOIEN	
		Receive underflow interrupt	SSISR.RUIRQ	SSICR.RUIEN	
		Receive overflow interrupt	SSISR.ROIRQ	SSICR.ROIEN	
		Idle state interrupt	SSISR.IIRQ	SSICR.IIEN	
	SSIRT11	Receive data full interrupt	SSIFSR.RDF	SSIFCR.RIE	Yes
		Transmit data empty interrupt	SSIFSR.TDE	SSIFCR.TIE	

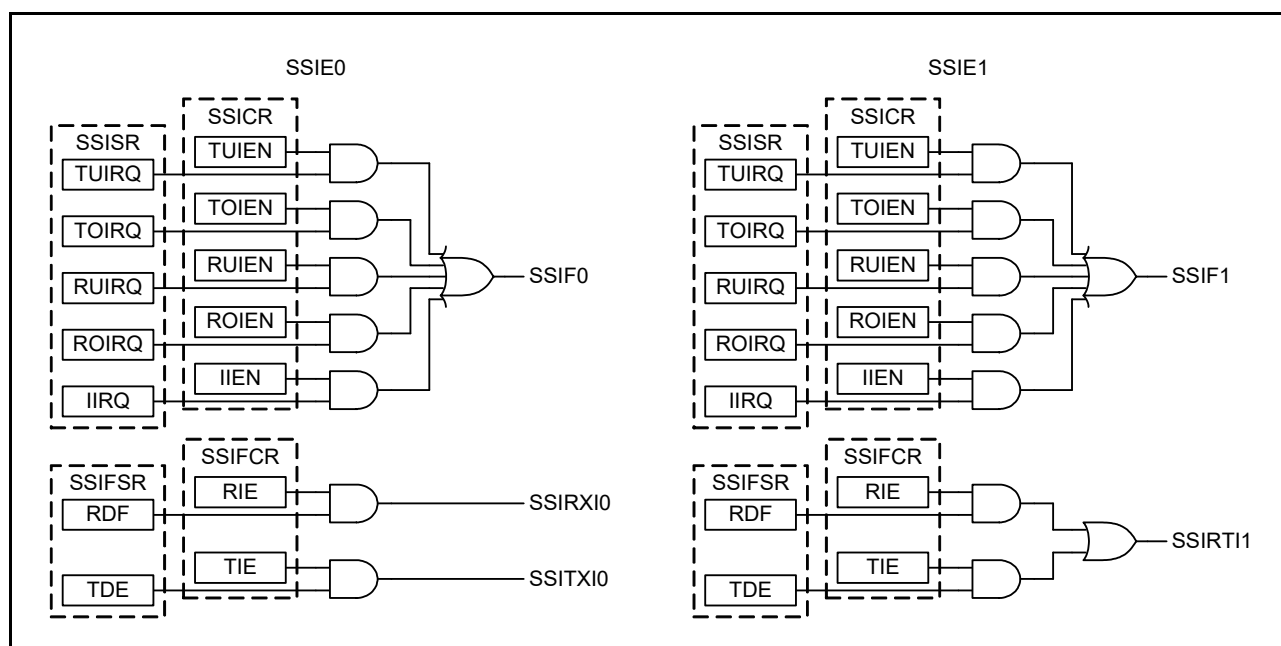


Figure 47.26 Interrupt System Block Diagram

47.4.1 SSIFn Interrupt

Each SSIFn interrupt combines five interrupt sources ($n = 0, 1$). Each interrupt source has a status flag and an interrupt enable bit, so the interrupt can be enabled or disabled individually.

If an SSIFn interrupt occurs, read the SSISR register in the interrupt handling routine to check the source of the interrupt request, and then perform proper operation. To clear the interrupt request, set the corresponding interrupt enable bit to 0 or clear the interrupt request flag.

47.4.2 SSITXIO Interrupt

The SSITXIO interrupt (channel 0 transmit data empty interrupt) is an edge detection interrupt. This interrupt request is generated when either of the following conditions is satisfied.

- The SSIFSR.TDE flag has changed from 0 to 1 while the SSIFCR.TIE bit is 1.
- The SSIFCR.TIE bit has been changed from 0 to 1 while the SSIFSR.TDE flag is 1.

Figure 47.27 shows a timing chart when an interrupt request is generated when the TDE flag changes.

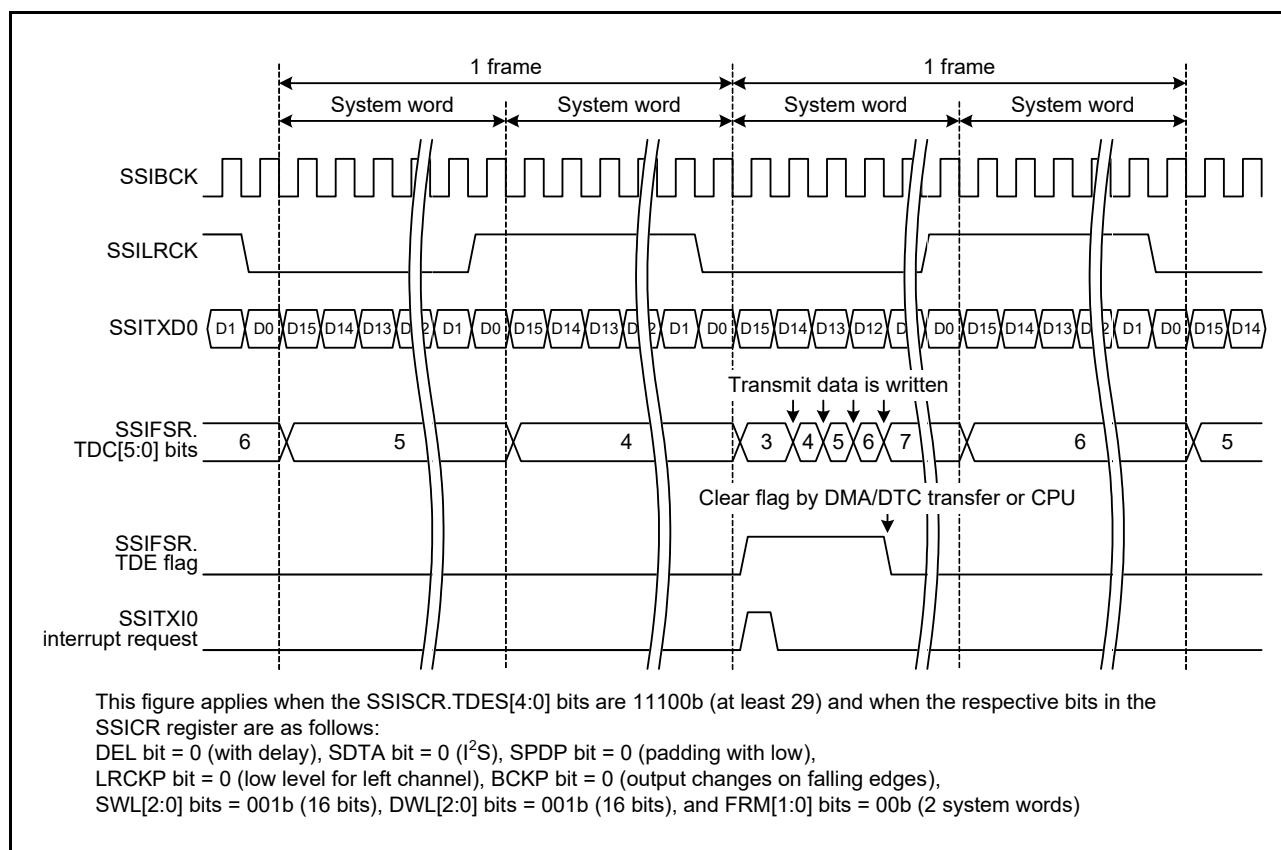


Figure 47.27 SSITXIO Interrupt Timing Chart

47.4.3 SSIRXI0 Interrupt

The SSIRXI0 interrupt (channel 0 receive data full interrupt) is an edge detection interrupt. This interrupt request is generated when either of the following conditions is satisfied.

- The SSIFSR.RDF flag has changed from 0 to 1 while the SSIFCR.RIE bit is 1.
- The SSIFCR.RIE bit has been changed from 0 to 1 while the SSIFSR.RDF flag is 1.

Figure 47.28 shows a timing chart when an interrupt request is generated when the RDF flag changes.

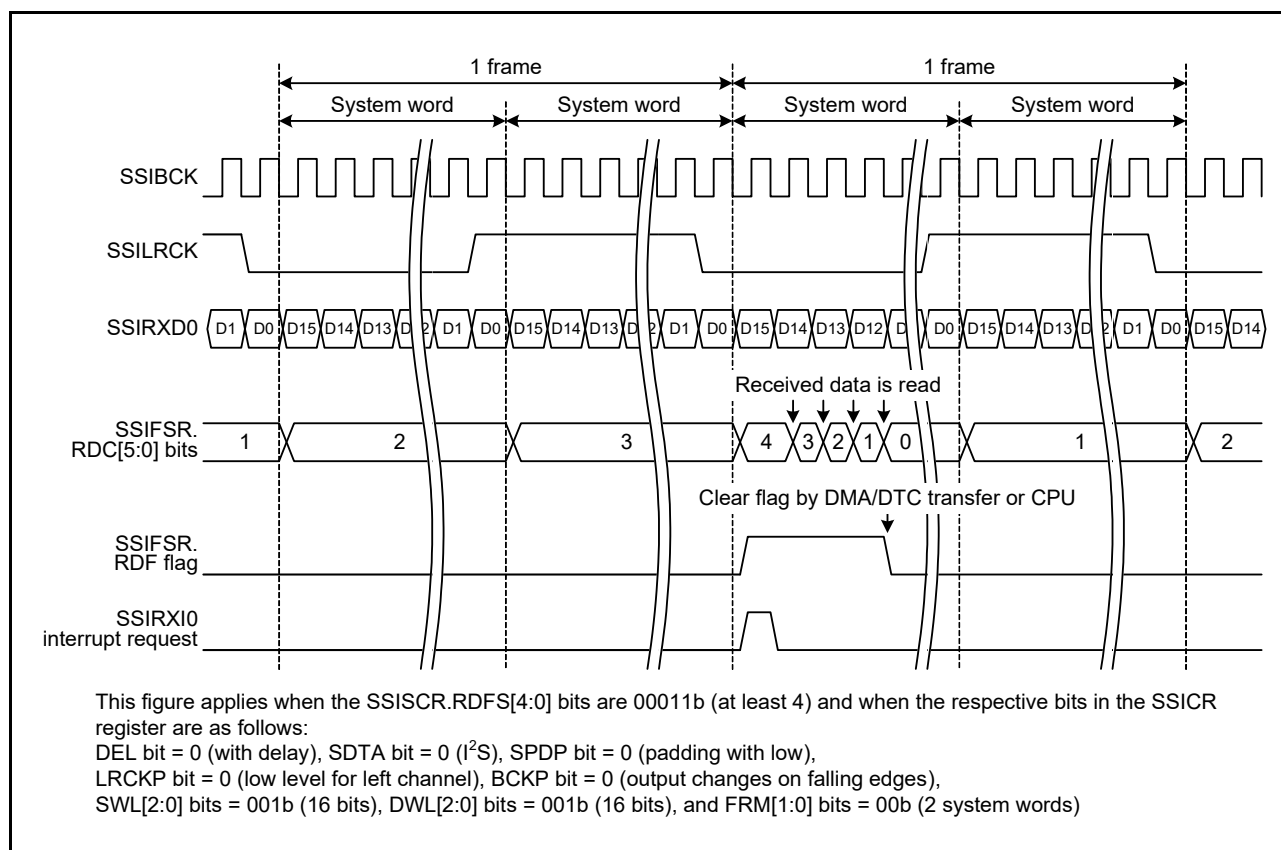


Figure 47.28 SSIRXI0 Interrupt Timing Chart

47.4.4 SSIRTI1 Interrupt

The operation of SSIRTI1 interrupt is the same as that of the SSITXI0 interrupt in transmission, and that of the SSIRXI0 interrupt in reception.

47.5 Notes

47.5.1 Setting the Module Stop Function

Operation of the SSIE can be enabled or disabled by setting a bit in the module stop control register D (MSTPCRD). The SSIE is initially disabled after a reset. Registers in the SSIE only become accessible after it has been released from the module stop state. For details, refer to [section 11, Low Power Consumption](#).

47.5.2 Note on LRCK in Slave Mode

The SSIE uses the SSILRCK signal as a start trigger in slave mode. Neither valid edges after the start trigger nor signal levels are used, so if a bit slippage occurs by a noise, the bit slippage cannot be repaired even at a frame boundary.